

UNIQUENESS OF GERVER'S SOFA

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ABSTRACT. The moving sofa problem asks for the largest area of a planar region that can be moved around the right-angled corner of a hallway of unit width. Gerver found a region of area $2.2195\dots$ in 1992 and conjectured that it is optimal; Baek proved this in 2024. We prove that Gerver's sofa is the only optimal region: every moving sofa of the same area is the image of Gerver's sofa under a rotation and a translation, as a set. Baek's proof studies one particular maximizer, produced by a compactness argument: its sofa, rotated, turns through a right angle, and at the right angle it satisfies an injectivity condition; a given maximizer need not be that one. We show that every maximizer has these properties, by approximating a maximizing convex cap by polygonal maximizers of a penalized problem, which are almost balanced, and passing to the limit. Equality in Baek's concave upper bound, whose nonlinear terms are areas swept by tangent segments, then forces differential equations for the difference of the support functions of the cap and of Gerver's cap, which identify the cap, and a last argument recovers the region from its cap. The proof, together with Baek's, is formalized in Lean 4 with Mathlib.

1. INTRODUCTION

1.1. The moving sofa problem. Let L be the L-shaped hallway of unit width, the union of the horizontal side $H_L = (-\infty, 1] \times [0, 1]$ and the vertical side $V_L = [0, 1] \times (-\infty, 1]$. A *moving sofa* is a connected planar region that can be moved rigidly from the horizontal side to the vertical side without leaving L (Definition 2.2). Moser [8] asked for the largest area of a moving sofa. A unit square has area 1. Hammersley [5] found a sofa of area $\pi/2 + 2/\pi = 2.2074\dots$, and Gerver [3] one of area $2.21953\dots$, bounded by 18 analytic arcs and segments, which he conjectured to be optimal. Romik [10] derived Gerver's sofa G from differential equations that express the local optimality of its sides, and described its boundary explicitly in terms of two angles that solve transcendental equations (Figure 1, Appendix A). Kallus and Romik [6] proved by a computer-assisted method that no sofa has area more than 2.37. Baek [1] proved Gerver's conjecture, in a paper of 119 pages: *every moving sofa has area at most $|G| = 2.2195\dots$*

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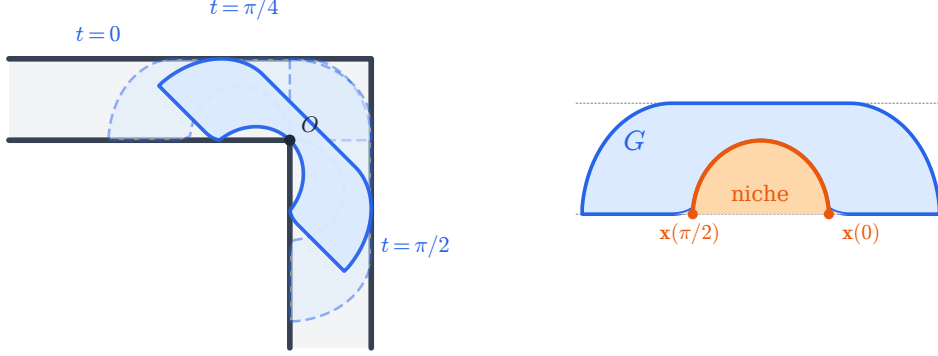


FIGURE 1. Gerver's sofa G in the hallway L . Left: it slides along the horizontal side (dashed), turns through a right angle around the inner corner (solid, halfway, turned by $\pi/4$) and leaves along the vertical side (dashed). Right: G with its niche, the hollow in its lower side that makes room for the corner, bounded by the path $\mathbf{x}$ of the inner corner of the hallway as seen from the sofa (orange).

1.2. Uniqueness. We prove that Gerver's sofa is the only optimal sofa.

Theorem 1.1 (uniqueness of Gerver's sofa). *Let S be a moving sofa with $|S| = |G|$. Then there are an angle θ and a vector v such that*

$$R_\theta S + v = G.$$

In other words, the moving sofas of maximum area are exactly the moving sofas that are rotated and translated copies of G (Corollary 9.5). Four remarks.

(i) The conclusion is an equality of sets, not of sets up to a null set. This uses that a moving sofa is closed and that G is the closure of its interior (Section 9); equality of areas alone would determine the set only up to a null set.

(ii) The formal statement allows a rotation, and the rotation is trivial: the width of G exceeds one in every direction other than the vertical, so that a copy $R_\theta G$ fits into the horizontal side of the hallway, where a moving sofa starts, only if θ is a multiple of π , and the proof of Theorem 1.1 produces $\theta \in [0, \pi/2 - \omega_0]$. Hence every moving sofa of maximal area is a translate of G (Remark 9.6; this refinement is not part of the formal statement). In particular not every rotated copy of G is a moving sofa.

(iii) Baek's paper does not claim uniqueness. Google DeepMind's formal-conjectures project lists the statement as an open problem [4] (at the time of writing), and we know of no earlier proof.

(iv) Along the way we obtain the following description of the extremal *caps*, which does not mention sofas: the caps with rotation angle $\pi/2$ that maximize

Baek's sofa area functional (Section 2.2) are the horizontal translates of Gerver's cap (Theorem 8.5).

1.3. Why Baek's argument does not give uniqueness, and what we do. Baek's proof has four steps. (1) After a translation, a moving sofa lies in a *monotone* sofa, which is a convex body K , its *cap*, minus a *niche* $\mathcal{N}(K)$ cut out by the inner corners of the supporting hallways; so the problem becomes the maximization of $\mathcal{A}_\omega(K) = |K| - |\mathcal{N}(K)|$ over caps with rotation angle ω . (2) For every ω he constructs a maximizing cap as a limit of *maximum polygon caps*, which are *balanced*: each side of the cap is as long as the corresponding side of the niche, since otherwise pushing one wall of one hallway would increase the area (Gerver's balancing argument). (3) From the balance he shows that, if the area is at least $11/5$, a rotated copy of the sofa $K \setminus \mathcal{N}(K)$ of this cap turns through the full right angle, so that only the rotation angle $\pi/2$ matters, and that for the rotation angle $\pi/2$ the cap satisfies the *injectivity condition*: the inner corner of the hallway, as seen from the sofa, moves steadily to the left and never crosses its own path. (4) For caps with the injectivity condition, a concave quadratic functional $\mathcal{Q}$ bounds the area, and Gerver's cap maximizes it.

Suppose now that S has area $|G|$. Every inequality of the chain becomes an equality, but the chain compares $|S|$ with the areas of *other* sets: of its monotonization, and of Baek's special balanced maximum sofa. A maximizing cap given in advance need not be a limit of maximum polygon caps, and then nothing is known about it: the properties of steps (2) and (3) are available only for the cap that was constructed.

We prove these properties for *every* maximizing cap. Let K_* be one. We approximate it by polygon caps K_n that maximize the polygon sofa area minus a penalty for straying from K_* (Section 4). The penalty is a weighted sum of squared differences of support values at dyadic normals, with weights that persist from one approximation to the next and add up to at most 1. Pushing one side of K_n as in Gerver's balancing argument shows that K_n is almost balanced: at each normal the difference between the length σ of the side of the cap and the length τ of the corresponding side of the niche is at most a multiple of the distance from K_n to K_* at the sampled normals, and by the choice of the weights these bounds add up to $O(d_H(K_n, K_*))$ (Section 5). In the limit this gives two inequalities between wedge gaps and side lengths of K_* (the pinned bounds) that suffice for Baek's rotation argument (Section 6), which produces a second monotone sofa U of area $|G|$ with rotation angle $\pi/2$ that contains a rigid image of S . For the maximizing cap K of U the same selection gives the inequality $\sigma_K \leq k_0(g_K^+(t))dt$ that gives the injectivity condition (Section 7). Then $|G| = \mathcal{A}(K) \leq \mathcal{Q}(\xi_K) \leq |G|$ is an equality, and equality in $\mathcal{Q}$ forces equality in each of its convex Mamikon terms, each the area swept by tangent segments, $\frac{1}{2} \int \varrho^2$ by Mamikon's theorem; this gives differential equations for the difference of the support functions of K and of Gerver's cap, whose solutions are $s \cos t$: K , and so U , is a horizontal

translate of Gerver’s cap and sofa (Section 8). Finally G is the closure of its interior, which turns the inclusion of a rigid image of S in G into an equality (Section 9).

What is new compared with [1] is Theorem 1.1 and its proof: the penalized selection with persistent samples, which gives the defect bounds for an arbitrary maximizer; the use of these bounds in place of the balance in Baek’s arguments of his Chapters 4 and 6; the equality case of Baek’s bound, with the description of the extremal caps (Theorem 8.5); and the regularity of G that recovers the set. The formalization of Theorem 1.1 is new; Baek’s theorem is also formalized in [9, 12], and twelve of the corrections to the statements of [1] (Appendix B) are due to the notes of [9].

1.4. Formal verification, and the use of AI. The proof, together with Baek’s, is formalized in Lean 4 with Mathlib [2, 13], in the repository [14]: every result of [1] used here, with the corrections of Appendix B, and the proof of Theorem 1.1. The statement of Theorem 1.1 is `Baek.gerver_sofa_unique` in `Challenge.lean`, and Appendix C lists the Lean declarations behind each result of this paper, with the exceptions noted there. Three of the Facts of Section 2 (Fact 2.15(a), Fact 2.16 and Fact 2.17) are known to us only through the formalization. The argument was first written, as an informal note, by an AI model (ChatGPT Pro 6) at the author’s request, and the formalization and this text were written by AI agents (Claude Opus 5.5 and Claude Sonnet 5.5) at the author’s request; Section 10 says exactly what each system did and what has and has not been checked. In particular, *the argument has not been peer reviewed, and no person has checked it independently of the formalization*; Lean’s kernel has checked the formalization.

1.5. Organization. Section 2 recalls the definitions and the results of [1] that we use. Section 3 reduces the theorem to caps and outlines the rest; Sections 4 to 9 follow the steps above; Section 10 describes the formalization and the provenance of the argument, and lists open questions. The appendices give the explicit description of Gerver’s sofa, the corrections to [1], and the dictionary to the Lean declarations.

2. SETTING, AND WHAT WE USE FROM BAEK’S PAPER

This section fixes the notation and records the results that the proof uses and does not prove; we call them *Facts*. Most are results of Baek’s paper [1], quoted in the form in which we use them and with the numbering of version 1 of the preprint (chapter.section.number). A few printed statements need a correction, and a few are used for objects slightly more general than those of their printed statements; each such place is noted with a label En , the number of the correction in the audit that accompanies the formalization [14], and Appendix B lists them. Some corrections come from the notes of another formalization of Baek’s proof [9]. Three Facts are known to us only through the formalization, because the printed proof is missing or invalid, or because

the statement is not in [1]: Fact 2.15(a) (E21), Fact 2.16 (E12, E24) and Fact 2.17; they are proved in [14], and Section 10 says what that means. The reader who knows [1] can skim this section and return to it for the notation.

2.1. Notation and convex bodies. The plane is $\mathbb{R}^2$, with points $p = (x, y)$, inner product $\langle p, q \rangle$ and norm $\|p\|$; $|X|$ is the area (Lebesgue measure) of $X \subset \mathbb{R}^2$. For an angle $t \in \mathbb{R}$ put

$$u_t = (\cos t, \sin t), \quad v_t = (-\sin t, \cos t) = u_{t+\pi/2},$$

and let R_t be the counterclockwise rotation by t . For $c \in \mathbb{R}$ let $l(t, c) = \{p : \langle p, u_t \rangle = c\}$, $H_-(t, c) = \{p : \langle p, u_t \rangle \leq c\}$, $H_+(t, c) = \{p : \langle p, u_t \rangle \geq c\}$, and let $H_\pm^\circ$ be the open half-planes. The half-plane $H_-(t, c)$ has *normal angle* t .

A *convex body* is a nonempty compact convex subset of $\mathbb{R}^2$; it may be a segment or a point. Its *support function* is $h_K(t) = \max_{p \in K} \langle p, u_t \rangle$, its *supporting line* is $l_K(t) = l(t, h_K(t))$, and $H_K(t) = H_-(t, h_K(t))$. Then $K = \bigcap_t H_K(t)$, so h_K determines K ; $h_{K+w}(t) = h_K(t) + \langle w, u_t \rangle$ and $h_{(1-\lambda)K+\lambda L} = (1-\lambda)h_K + \lambda h_L$ for Minkowski combinations. The Hausdorff distance is $d_H(K, L) = \sup_t |h_K(t) - h_L(t)|$, and $K_n \rightarrow K$ means $d_H(K_n, K) \rightarrow 0$. The *edge* of K with normal angle t is the segment $e_K(t) = K \cap l_K(t) = [v_K^-(t), v_K^+(t)]$, where $v_K^+(t)$ is its endpoint farthest in the direction v_t ; it may be a point. For angles a, b with $\sin(b-a) \neq 0$ the supporting lines $l_K(a)$ and $l_K(b)$ meet at

$$v_K(a, b) = h_K(a) u_a + \frac{h_K(b) - h_K(a) \cos(b-a)}{\sin(b-a)} v_a.$$

Fact 2.1 (convex bodies). *Let K be a convex body.*

- (a) h_K has the one-sided derivatives $h'_K(t^\pm) = \langle v_K^\pm(t), v_t \rangle$; v_K^+ is right-continuous and v_K^- left-continuous; $v_K(t, s) \rightarrow v_K^+(t)$ as $s \downarrow t$ and $v_K(s, t) \rightarrow v_K^-(t)$ as $s \uparrow t$. [1, Thm. 2.1.3]
- (b) The surface area measure σ_K is the Lebesgue-Stieltjes measure on $\mathbb{R}$ of $G_K(t) = \langle v_K^+(t), v_t \rangle + \int_0^t h_K$ (formally $\sigma_K = h''_K + h_K$); it is 2π -periodic. We write $\sigma_K(t) = \sigma_K(\{t\})$. Its atoms are the edge lengths: $\sigma_K(t) = |e_K(t)|$ and $v_K^+(t) = v_K^-(t) + \sigma_K(t) v_t$. Moreover $v_K^+(d) - v_K^+(c) = \int_{(c,d]} v_s d\sigma_K(s)$ for $c \leq d$, and $|K| = \frac{1}{2} \int_{[0,2\pi)} h_K d\sigma_K$. [1, Prop. 2.1.2, Thm. 5.2.2], [11, Rem. 5.1.2]
- (c) Every sequence of convex bodies in a bounded set has a subsequence that converges to a convex body (Blaschke's selection theorem), and $K \mapsto |K|$ is continuous for d_H . [11, Section 1.8, in particular Thm. 1.8.20]

Baek's Thm. 2.1.3 states the limits in (a); the formulas for the one-sided derivatives follow from them and from $\langle v_K(t, s), v_t \rangle = (h_K(s) - h_K(t) \cos(s-t)) / \sin(s-t)$, and the printed statement has a misprint (E27). From (b) one reads off, for $a < b$,

$$\sigma_K((a, b)) = \langle v_K^-(b), v_b \rangle - \langle v_K^+(a), v_a \rangle + \int_a^b h_K. \quad (2.1)$$

2.2. Moving sofas, monotoneization, caps and niches. The *hallway* is $L = H_L \cup V_L$, the union of its horizontal side $H_L = (-\infty, 1] \times [0, 1]$ and its vertical side $V_L = [0, 1] \times (-\infty, 1]$; its inner corner is the origin O and its outer corner is $(1, 1)$.

Definition 2.2 (moving sofa). A *movement* of a set $S \subset \mathbb{R}^2$ is a pair of continuous maps $\theta : [0, 1] \rightarrow \mathbb{R}$, $c : [0, 1] \rightarrow \mathbb{R}^2$ with $\theta(0) = 0$ such that the rigid motions $\Phi_s(p) = R_{\theta(s)}p + c(s)$ satisfy

$$\Phi_0(S) \subset H_L, \quad \Phi_s(S) \subset L \quad (0 \leq s \leq 1), \quad \Phi_1(S) \subset V_L.$$

Its *rotation angle* is $\omega = -\theta(1)$. A *moving sofa* is a nonempty, closed and connected set that has a movement; it is a moving sofa *with rotation angle* ω if it has a movement with that rotation angle.

This is [1, Def. 1.1.2]: Baek moves the set along a continuous curve of orientation-preserving isometries that starts at a translation, and the angle of such a curve lifts to a continuous real function. A moving sofa is bounded, hence compact: the start and end positions bound it unless $\theta(1) \in \pi/2 + 2\pi\mathbb{Z}$, and then a position in the middle of the movement does (in Lean, `isBounded_of_isMovingSofa`). A translate of a moving sofa with rotation angle ω is again one. Let $\omega_0 = \sec^{-1}(11/5) = \arccos(5/11) = 1.0989 \dots$ (63° approximately).

For $\omega \in (0, \pi/2]$ let $H = \mathbb{R} \times [0, 1]$ be the horizontal strip, $V_\omega = \{p : 0 \leq \langle p, u_\omega \rangle \leq 1\}$, $P_\omega = H \cap V_\omega$ the parallelogram (a strip if $\omega = \pi/2$), and $F_\omega = \{p : y \geq 0, \langle p, u_\omega \rangle \geq 0\}$ the *fan*. If $\omega < \pi/2$ the two upper sides of P_ω meet at $o_\omega = (c_\omega, 1)$, $c_\omega = \sec \omega - \tan \omega$, and the two lower sides at O . A moving sofa S with rotation angle ω is in *standard position* if $h_S(\omega) = h_S(\pi/2) = 1$.

Fact 2.3 (rotation angle and standard position). (a) *Every moving sofa S with $|S| \geq 11/5$ is a moving sofa with some rotation angle $\omega \in [\omega_0, \pi/2]$.* [1, Thm. 1.5.1]

(b) *If S is a moving sofa with rotation angle $\omega \in (0, \pi/2]$, some translate $S + v$ is in standard position, and a moving sofa with rotation angle ω in standard position lies in P_ω .* [1, Prop. 1.2.1, Prop. 2.3.1]

Supporting hallways. For a compact set S and an angle t let

$$f_{S,t}(p) = R_t p + (h_S(t) - 1)u_t + (h_S(t + \pi/2) - 1)v_t,$$

and let $L_S(t) = f_{S,t}(L)$ be the *supporting hallway* of S at t : the translate of $R_t L$ whose two outer walls $a_S(t) = l_S(t)$ and $c_S(t) = l_S(t + \pi/2)$ are supporting lines of S . Its inner corner and outer corner are

$$\begin{aligned} \mathbf{x}_S(t) &= (h_S(t) - 1)u_t + (h_S(t + \pi/2) - 1)v_t, \\ \mathbf{y}_S(t) &= h_S(t)u_t + h_S(t + \pi/2)v_t = \mathbf{x}_S(t) + u_t + v_t, \end{aligned}$$

its inner walls lie on the lines $b_S(t) = l(t, h_S(t) - 1)$ and $d_S(t) = l(t + \pi/2, h_S(t + \pi/2) - 1)$ (the half-lines from $\mathbf{x}_S(t)$ in the directions $-v_t$ and $-u_t$ are $\vec{b}_S(t)$ and $\vec{d}_S(t)$), and

$$\begin{aligned} L_S(t) &= Q_S^+(t) \setminus Q_S^-(t), & Q_S^+(t) &= H_S(t) \cap H_S(t + \frac{\pi}{2}), \\ Q_S^-(t) &= H_-^\circ(t, h_S(t) - 1) \cap H_-^\circ(t + \frac{\pi}{2}, h_S(t + \frac{\pi}{2}) - 1). \end{aligned}$$

$Q_S^-(t)$ is the open quarter-plane below the inner corner. A moving sofa in standard position lies in $L_S(t)$ for every $t \in [0, \omega]$, and the *monotonization* of S is

$$\mathcal{I}(S) = P_\omega \cap \bigcap_{t \in [0, \omega]} L_S(t).$$

A *monotone sofa* with rotation angle ω is a set $\mathcal{I}(S')$ for a moving sofa S' in standard position with rotation angle ω .

Caps and niches. Let $J_\omega = [0, \omega] \cup [\pi/2, \omega + \pi/2]$. A *cap* with rotation angle $\omega \in (0, \pi/2]$ is a convex body K with

$$h_K(\omega) = h_K(\pi/2) = 1, \quad h_K(\omega + \pi) = h_K(3\pi/2) = 0,$$

that is an intersection of closed half-planes with normal angles in $J_\omega \cup \{\omega + \pi, 3\pi/2\}$. The caps with rotation angle ω form the set $\mathcal{K}_\omega^c$. A cap lies in P_ω and touches its four sides; and

$$K = F_\omega \cap \bigcap_{s \in J_\omega} H_K(s). \quad (2.2)$$

The *niche* of a cap K and its *sofa area* are

$$\mathcal{N}(K) = F_\omega \cap \bigcup_{t \in (0, \omega)} Q_K^-(t), \quad \mathcal{A}_\omega(K) = |K| - |\mathcal{N}(K)|.$$

For S in standard position, the *cap* of S is $\mathcal{C}(S) = P_\omega \cap \bigcap_{t \in [0, \omega]} Q_S^+(t)$. The reflection $M_\omega(x, y) = (-x \sin \omega + y \cos \omega, x \cos \omega + y \sin \omega)$ in the line through O at the angle $\pi/4 + \omega/2$ (the line through O and o_ω if $\omega < \pi/2$) maps P_ω and F_ω to themselves; for $\omega = \pi/2$ it is $(x, y) \mapsto (-x, y)$. The *mirror image* of a cap K is $K^m = M_\omega(K)$.

Fact 2.4 (monotone sofas and caps). *Let $\omega \in (0, \pi/2]$.*

- (a) *If S is a moving sofa with rotation angle ω in standard position, then $\mathcal{I}(S)$ is a moving sofa with rotation angle ω in standard position and $S \subset \mathcal{I}(S)$. [1, Thm. 2.3.2]*
- (b) *For such an S , $K = \mathcal{C}(S)$ is a cap with $h_K = h_S$ on J_ω , and $\mathcal{I}(S) = K \setminus \mathcal{N}(K)$. [1, Lemma 2.3.5, Thm. 2.4.1, Thm. 2.4.2]*
- (c) *If T is a monotone sofa with rotation angle ω and $K = \mathcal{C}(T)$, then $T = K \setminus \mathcal{N}(K)$, $\mathcal{N}(K) \subset K$ and $|T| = \mathcal{A}_\omega(K)$. [1, Thm. 2.4.3, Thm. 2.5.9, Thm. 2.5.10]*
- (d) *If K is a cap, so is K^m , with $h_{K^m}(t) = h_K(\omega + \pi/2 - t)$, $\mathcal{N}(K^m) = M_\omega(\mathcal{N}(K))$, $\mathcal{A}_\omega(K^m) = \mathcal{A}_\omega(K)$ and $\sigma_{K^m}(E) = \sigma_K(\omega + \pi/2 - E)$ for Borel sets E . [1, Prop. 2.5.4]*

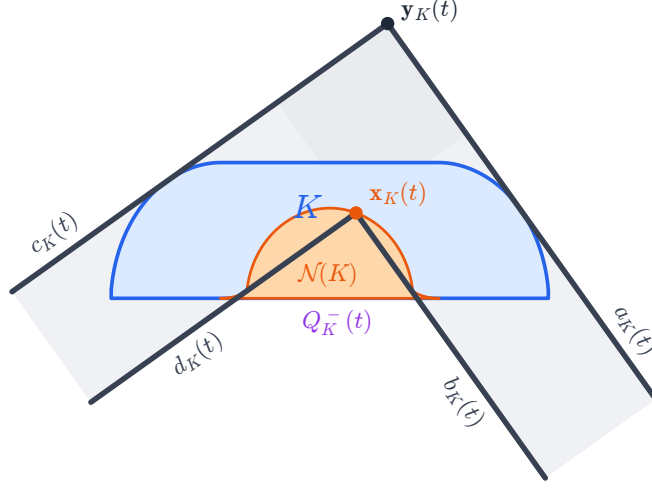


FIGURE 2. A cap, its niche and a supporting hallway: Gerver's cap $K = K_G$ (blue), its niche $\mathcal{N}(K)$ (orange), and the supporting hallway $L_K(t)$ at $t = 0.62$ (grey), whose outer walls $a_K(t)$ and $c_K(t)$ touch K and meet at the outer corner $\mathbf{y}_K(t)$, and whose inner walls $b_K(t)$ and $d_K(t)$ meet at the inner corner $\mathbf{x}_K(t)$, on the boundary of the niche. $Q_K^-(t)$ is the open quarter-plane below the inner corner; the niche is the union of these quarter-planes over $t \in (0, \pi/2)$, cut at the x -axis.

Three places need a remark. The proof of (a) uses Def. 2.3.9, which defines the left and the right side of a line for lines not parallel to the y -axis, where it needs the x -axis (E1). In (b), the union over $t \in [0, \omega]$ in Thm. 2.4.2 and the union over $(0, \omega)$ in the definition of the niche agree after intersection with F_ω , and $\mathcal{C}(S)$ is bounded also for $\omega = \pi/2$, which Baek does not show (E2). In (d), Prop. 2.5.4 prints $?_{K^m}(t) = M_\omega(?_K(\omega - t))$ for each of the walls a, b, c, d and the ends W, Z , whereas M_ω exchanges $a \leftrightarrow c$, $b \leftrightarrow d$ and $W \leftrightarrow Z$, and the identity for the vertices has K^m in place of K on the right (E3); (d) uses none of this.

Two more facts about caps are used without proof in Baek's [1, Lemma 2.5.6], [1, Lemma 3.4.8] and [1, Thm. 2.5.8] (E4): that O and o_ω lie in every cap if $\omega < \pi/2$ (the statement about o_ω is not in the audit), and where the ends of the *upper boundary* $\bigcup_{t \in [0, \omega + \pi/2]} e_K(t)$ of a cap K are. Here $A_K^\pm(t) = v_K^\pm(t)$ and $C_K^\pm(t) = v_K^\pm(t + \pi/2)$ [1, Def. 2.5.1, Def. 2.5.2].

Lemma 2.5 (every cap contains O and o_ω). *If $\omega < \pi/2$, every cap $K \in \mathcal{K}_\omega^c$ contains O and o_ω ; in particular $h_K \geq 0$. For $\omega = \pi/2$ the origin need not lie in K (take $K = [5, 6] \times [0, 1]$).*

Proof. Let $q \in K \cap l(\pi/2, 1)$ and $q' \in K \cap l(\omega, 1)$. As $K \subset P_\omega$, $q = o_\omega - \lambda u_0$ and $q' = o_\omega - \mu v_\omega$ with $\lambda, \mu \geq 0$. For $s \in [0, \omega]$, $\langle o_\omega, u_s \rangle = \langle q', u_s \rangle + \mu \sin(s - \omega) \leq \langle q', u_s \rangle \leq h_K(s)$, and for $s \in [\pi/2, \omega + \pi/2]$, $\langle o_\omega, u_s \rangle = \langle q, u_s \rangle + \lambda \cos s \leq \langle q, u_s \rangle \leq h_K(s)$. As $o_\omega \in F_\omega$, (2.2) gives $o_\omega \in K$. Then $h_K(s) \geq \langle o_\omega, u_s \rangle = c_\omega \cos s + \sin s = \sqrt{1 + c_\omega^2} \sin(s + \frac{\pi}{4} - \frac{\omega}{2})$, because $c_\omega = \tan(\pi/4 - \omega/2)$, and $0 < s + \pi/4 - \omega/2 < \pi$ for $s \in J_\omega$, so $h_K \geq 0$ on J_ω . Hence $O \in F_\omega$ satisfies $\langle O, u_s \rangle = 0 \leq h_K(s)$ for $s \in J_\omega$, and $O \in K$ by (2.2); and $h_K(s) \geq \langle O, u_s \rangle = 0$ for every s . $\square$

Lemma 2.6 (the ends of the upper boundary). *For every cap $K \in \mathcal{K}_\omega^c$,*

$$A_K^-(0) = (h_K(0), 0) \quad \text{and} \quad C_K^+(\omega) = h_K(\omega + \pi/2) v_\omega; \quad (2.3)$$

they lie on $l(\pi/2, 0)$ and on $l(\omega, 0)$.

Proof. Let $q \in e_K(\omega + \pi/2)$ and $p = q - \langle q, u_\omega \rangle u_\omega = h_K(\omega + \pi/2) v_\omega$. Since $\langle q, u_\omega \rangle \geq 0$ ($K \subset P_\omega$) and $\cos(s - \omega) \geq 0$ for $s \in J_\omega$, we have $\langle p, u_s \rangle \leq \langle q, u_s \rangle \leq h_K(s)$ for $s \in J_\omega$. Also $p \in F_\omega$, since $h_K(\omega + \pi/2) \geq 0$ (Lemma 2.5; for $\omega = \pi/2$, $p_y = 0$). So $p \in K$ by (2.2). As $\langle \cdot, u_\omega \rangle \geq 0 = \langle p, u_\omega \rangle$ on K , p is the point of $e_K(\omega + \pi/2)$ farthest in the direction $v_{\omega+\pi/2} = -u_\omega$, that is $p = C_K^+(\omega)$. The mirror image K^m is a cap (Fact 2.4(d)) with $h_{K^m}(\omega + \pi/2) = h_K(0)$, and M_ω maps v_ω to u_0 and reverses orientation, so that $C_{K^m}^+(\omega) = h_K(0) v_\omega$ is mapped to the lower end $A_K^-(0)$ of $e_K(0)$, which is $(h_K(0), 0)$. $\square$

The next fact is the main theorem of [1], in the form for caps. It assembles Baek's construction of a balanced maximum cap K_ω for every ω ([1, Thm. 3.5.2, Thm. 3.5.4]), its maximality ([1, Thm. 3.5.5]), and the optimality of Gerver's sofa G ([1, Thm. 1.1.1]), which is recalled in Section 2.6.

Fact 2.7 (optimality). *Every moving sofa S has $|S| \leq |G|$, and $|G| > 11/5$. For every $\omega \in (0, \pi/2]$ and every cap $K \in \mathcal{K}_\omega^c$, $\mathcal{A}_\omega(K) \leq |G|$.*

For the last claim, [1, Thm. 3.5.6] gives a monotone sofa $S_\omega = K_\omega \setminus \mathcal{N}(K_\omega)$ whose cap K_ω satisfies $\mathcal{A}_\omega(K) \leq \mathcal{A}_\omega(K_\omega)$ for every cap K [1, Thm. 3.5.5], and $\mathcal{A}_\omega(K_\omega) = |S_\omega| \leq |G|$ by [1, Thm. 1.1.1] and Fact 2.4(c).

2.3. Polygon caps. Let $\omega \in (0, \pi/2]$. An *angle set* is a finite nonempty $\Theta \subset (0, \omega)$, and $\Theta^\diamond = \Theta \cup (\Theta + \pi/2) \cup \{\omega, \pi/2\} \subset (0, \pi)$ is its set of *defining normals*; the two normals ω and $\pi/2$, which carry the two strips of P_ω , are *pinned*, the others *floating*. A *polygon cap* with angle set Θ is a cap that is an intersection of closed half-planes with normal angles in $\Theta^\diamond \cup \{\omega + \pi, 3\pi/2\}$; they form the set $\mathcal{K}_\Theta^c \subset \mathcal{K}_\omega^c$. A polygon cap is the intersection of its supporting half-planes $H_K(s)$ at these normal angles. For a cap K put

$$\begin{aligned} \mathcal{C}_\Theta(K) &= P_\omega \cap \bigcap_{t \in \Theta} Q_K^+(t), & \mathcal{N}_\Theta(K) &= F_\omega \cap \bigcup_{t \in \Theta} Q_K^-(t), \\ \mathcal{A}_\Theta(K) &= |\mathcal{C}_\Theta(K)| - |\mathcal{N}_\Theta(K)|. \end{aligned}$$

Here the *fan* F_ω , and not the parallelogram P_ω , cuts the polygon niche: the printed [1, Def. 3.2.5] has P_ω and with it [1, Prop. 3.3.5] and [1, Lemma 3.4.2] are false, whereas the results of [1, Ch. 3] hold with F_ω , as the formalization proves (E6).

Fact 2.8 (the polygon bound). *For every cap K and angle set Θ , $\mathcal{C}_\Theta(K)$ is a polygon cap that contains K and has the same support values as K on Θ° , and $\mathcal{C}_\Theta(K) = K$ if $K \in \mathcal{K}_\Theta^\circ$; $\mathcal{N}_\Theta(K) = \mathcal{N}_\Theta(\mathcal{C}_\Theta(K)) \subset \mathcal{N}(K)$. Hence $\mathcal{A}_\omega(K) \leq \mathcal{A}_\Theta(K) = \mathcal{A}_\Theta(\mathcal{C}_\Theta(K))$, and $\mathcal{A}_\Theta(K) = |K| - |\mathcal{N}_\Theta(K)|$ for a polygon cap. [1, Prop. 3.2.1, Prop. 3.2.2, Thm. 3.2.3]*

Fact 2.9 (bounded width). *For $t \in (0, \omega)$ there is a constant $c_{\omega, t}$ such that every polygon cap K with an angle set $\Theta \ni t$ and $\mathcal{A}_\Theta(K) > 0$ has width $h_K(0) + h_K(\pi) \leq c_{\omega, t}$. [1, Lemma 3.4.2]*

Sides of the polygon niche. For a polygon cap K with angle set Θ , the boundary of $F_\omega \setminus \mathcal{N}_\Theta(K)$ consists of an open half-line, a polyline $\mathbf{p}_K$ from $C_K^+(\omega)$ to $A_K^-(0)$ (see (2.3)), along which the abscissa increases and whose segments have normal angles in Θ° , and an open half-line [1, Thm. 3.4.4]. For $t \in \Theta^\circ$ let $\tau_K(t)$ be the total length of the segments of $\mathbf{p}_K$ with normal angle t , and $\sigma_K(t) = |e_K(t)|$ the length of the edge of K with normal angle t . The following is about every polygon cap; maximality plays no role in it.

Fact 2.10 (sides of the polygon niche). *Let $K \in \mathcal{K}_\Theta^\circ$.*

- (a) *For $t \in \Theta$, $\mathcal{H}^1(\partial \mathcal{N}_\Theta(K) \cap \vec{b}_K(t)) = \tau_K(t)$ and $\mathcal{H}^1(\partial \mathcal{N}_\Theta(K) \cap \vec{d}_K(t)) = \tau_K(t + \pi/2)$. [1, Lemma 3.4.5 (1)]*
- (b) *For $t \in \{\omega, \pi/2\}$, $\mathcal{H}^1(\mathcal{N}_\Theta(K) \cap l(t, 0)) = \sigma_K(t + \pi) - \tau_K(t)$. [1, Lemma 3.4.5 (2)]*
- (c) $\sum_{t \in \Theta^\circ} \sin t (\sigma_K(t) - \tau_K(t)) = 0$. [1, Proof of Lemma 3.4.6]

Baek obtains (c) from the vector identity $\sum_t \tau_K(t) v_t = C_K^+(\omega) - A_K^-(0) = \sum_t \sigma_K(t) v_t$, by taking the inner product with u_0 (note $\langle v_t, u_0 \rangle = -\sin t$); the formalization proves (c) in this sine form, and of the vector identity only its σ half.

Support values and first variations. For $h : \Theta^\circ \rightarrow \mathbb{R}$ let

$$\begin{aligned}
 P_h &= \bigcap_{s \in \{\omega, \pi/2\}} (H_-(s, h(s)) \cap H_+(s, h(s) - 1)), \\
 \mathcal{C}_\Theta(h) &= P_h \cap \bigcap_{s \in \Theta^\circ \setminus \{\omega, \pi/2\}} H_-(s, h(s)), \\
 F_h &= \bigcap_{s \in \{\omega, \pi/2\}} H_+(s, h(s) - 1), \\
 \mathcal{N}_\Theta(h) &= F_h \cap \bigcup_{t \in \Theta} (H_-^\circ(t, h(t) - 1) \cap H_-^\circ(t + \frac{\pi}{2}, h(t + \frac{\pi}{2}) - 1)),
 \end{aligned}$$

and $\mathcal{A}_\Theta(h) = |\mathcal{C}_\Theta(h)| - |\mathcal{N}_\Theta(h)|$ ([1, Def. 3.3.3]). For a polygon cap K and $h = h_K|_{\Theta^\diamond}$ these are K , $\mathcal{N}_\Theta(K)$ and $\mathcal{A}_\Theta(K)$ ([1, Prop. 3.3.4, Prop. 3.3.5], with F_ω , E6). For $\varepsilon > 0$ and $t \in \Theta^\diamond$ let $h_K^{t,\varepsilon}$ be h_K with the value at t replaced by $h_K(t) + \varepsilon$.

Fact 2.11 (pushing one side). *Let $K \in \mathcal{K}_\Theta^c$ and $t \in \Theta^\diamond$.*

(a) *There are $\varepsilon_0 > 0$ and $M \geq 0$ such that for $0 < \varepsilon \leq \varepsilon_0$*

$$|\mathcal{A}_\Theta(h_K^{t,\varepsilon}) - \mathcal{A}_\Theta(K) - (\sigma_K(t) - \tau_K(t))\varepsilon| \leq M\varepsilon^2.$$

(b) *If $\sigma_K(t) > 0$, then for small $\varepsilon > 0$ the set $\mathcal{C}_\Theta(h_K^{t,\varepsilon})$ is a translate $C + v$ of a polygon cap $C \in \mathcal{K}_\Theta^c$.*

(c) *If $\mathcal{C}_\Theta(h_K^{t,\varepsilon}) = C + v$ with $C \in \mathcal{K}_\Theta^c$, then $\mathcal{A}_\Theta(h_K^{t,\varepsilon}) \leq \mathcal{A}_\Theta(C)$.*

[1, Lemma 3.4.7, Lemma 3.4.8, Prop. 3.3.7, Thm. 3.3.6]

Part (a) rests on [1, Thm. 3.1.2], which needs the Nef polygon (a set obtained from finitely many half-planes by intersections, unions and complements) to be bounded; every use is to bounded polygons (E5, and Section 4 of REPORT.md in [14]). In the proof of [1, Lemma 3.4.7] the sign of the change of $|\mathcal{N}_\Theta|$ is misprinted (E27).

2.4. The injectivity condition and arm lengths. The rest of this section, and Sections 7 and 8, concern caps with rotation angle $\pi/2$, and $\mathcal{A} = \mathcal{A}_{\pi/2}$. For $K \in \mathcal{K}_{\pi/2}^c$ the *arm lengths* are

$$f_K^\pm(t) = \langle \mathbf{y}_K(t) - A_K^\pm(t), v_t \rangle, \quad g_K^\pm(t) = \langle \mathbf{y}_K(t) - C_K^\pm(t), u_t \rangle,$$

with $\mathbf{y}_K(t) = h_K(t)u_t + h_K(t + \pi/2)v_t$ the outer corner of the supporting hallway and $A_K^\pm, C_K^\pm$ as in (2.3). Thus $\mathbf{y}_K(t) = A_K^\pm(t) + f_K^\pm(t)v_t = C_K^\pm(t) + g_K^\pm(t)u_t$: these are the distances from the outer corner to the points where K touches the two outer walls. They are nonnegative, $g_K^- \leq g_K^+$, and $f_K^\pm, g_K^\pm$ are at most the diameter of K , because $A_K^\pm(t) - C_K^\pm(t) = g_K^\pm(t)u_t - f_K^\pm(t)v_t$ joins two points of K .

Definition 2.12 (injectivity condition). A cap $K \in \mathcal{K}_{\pi/2}^c$ satisfies the *injectivity condition* if

- (1) σ_K has a density on $[0, \pi/2)$ and on $(\pi/2, \pi]$ (the top edge, at $\pi/2$, may carry an atom);
- (2) the inner corner $\mathbf{x}_K$ is continuously differentiable on $[0, \pi/2]$;
- (3) $\langle \mathbf{x}'_K(t), u_t \rangle < 0 < \langle \mathbf{x}'_K(t), v_t \rangle$ for $0 < t < \pi/2$.

We write $\mathcal{K}^i$ for the set of caps in $\mathcal{K}_{\pi/2}^c$ that satisfy the injectivity condition and have $|K| \geq 11/5$.

Condition (3) implies that the inner corner moves strictly to the left, so the path of the inner corner has no self-intersections. We use the following.

Fact 2.13 (arms). *Let $K \in \mathcal{K}_{\pi/2}^c$.*

- (a) For every t , $\mathbf{x}_K$ has the right derivative $-(f_K^+(t) - 1)u_t + (g_K^+(t) - 1)v_t$, and the left derivative given by the same formula with f_K^-, g_K^- . [1, Thm. 6.2.3]
- (b) $f_K^+(b) - f_K^+(a) = \int_a^b g_K^+ - \sigma_K((a, b])$ for $0 \leq a \leq b \leq \pi/2$. [1, Thm. 6.2.5]
- (c) For the reflection K^m in $x \mapsto -x$, $f_{K^m}^\pm(t) = g_K^\mp(\pi/2 - t)$ and $g_{K^m}^\pm(t) = f_K^\mp(\pi/2 - t)$. [1, Prop. 6.2.2]
- (d) If K satisfies (1), then $A_K^+ = A_K^-$ and $f_K^+ = f_K^-$ on $[0, \pi/2)$, $C_K^+ = C_K^-$ and $g_K^+ = g_K^-$ on $(0, \pi/2]$. With $A_K = A_K^-$, $C_K = C_K^+$, $f_K = f_K^-$ and $g_K = g_K^+$ these four functions are continuous on $[0, \pi/2]$; $f_K(0) = g_K(\pi/2) = 1$, and $\mathbf{x}'_K(t) = -(f_K(t) - 1)u_t + (g_K(t) - 1)v_t$. In particular (2) holds, and (3) says that $f_K, g_K > 1$ on $(0, \pi/2)$. [1, Def. 6.4.1, Prop. 6.4.5, Prop. 6.4.6] The end values $f_K(0) = g_K(\pi/2) = 1$ are not stated there (E17).

In (c) the printed statement forgets that the reflection reverses the angle (E13); in (a) the right derivative is called the left one, and the second display of the printed theorem is about the left derivative (E14). Let

$$k_0(x) = \max\left(|x - 1|, \frac{|x-1|+1}{2}\right), \quad m_0(x) = x - k_0(x).$$

The function k_0 is 1-Lipschitz, and m_0 is nondecreasing, with $m_0(x) = \frac{3}{2}x - 1$ on $[0, 1]$, $m_0(x) = \frac{x}{2}$ on $[1, 2]$ and $m_0(x) = 1$ on $[2, \infty)$. For a continuous f on $[0, \pi/2]$ let $\mathcal{F}f(x) = 1 + \int_0^x m_0(f(\pi/2 - u))du$, and let $f_0 = 0$, $f_{j+1} = \max(f_j, \mathcal{F}f_j)$.

Fact 2.14 (convergence of arms and the iteration). (a) If polygon caps K_n with rotation angle $\pi/2$ converge in d_H to a cap $K \in \mathcal{K}_{\pi/2}^c$, then

$$\int_0^{\pi/2} |g_{K_n}^+ - g_K^+| \rightarrow 0. \quad [1, \text{Lemma 6.4.2}]$$

- (b) $f_{11}(x) > 1$ for $x \in (0, \pi/2]$. [1, Lemma 6.5.5], which is printed for $x \in (0, 1]$ and used on $(0, \pi/2]$ (E17)

2.5. Baek's upper bound. Let $\varphi = \varphi_G = 0.03917\dots$ be Gerver's angle (Section 2.6), $\varphi^R = \varphi$ and $\varphi^L = \pi/2 - \varphi$. For $K \in \mathcal{K}_{\pi/2}^c$ let $H_K^b(t) = \{\langle p, u_t \rangle \geq h_K(t) - 1\}$ and $H_K^d(t) = \{\langle p, v_t \rangle \geq h_K(t + \pi/2) - 1\}$ be the closed half-planes bounded by the inner walls $b_K(t)$ and $d_K(t)$, on the side away from $Q_K^-(t)$, and

$$B_K = K \cap \bigcap_{t \in [\varphi^R, \pi/2]} H_K^b(t), \quad D_K = K \cap \bigcap_{t \in [0, \varphi^L]} H_K^d(t).$$

Baek's upper bound is the functional $\mathcal{Q}$ of [1, Def. 8.2.2] on the set $\mathcal{T}$ of triples (K, B, D) of convex bodies with $K \in \mathcal{K}^i$, $B, D \subset K$, $h_K(t) + h_B(\pi + t) \leq 1$ on $[\varphi^R, \pi/2]$ with equality at the ends, and $h_K(\pi/2 + t) + h_D(3\pi/2 + t) \leq 1$ on $[0, \varphi^L]$ with equality at the ends. The sets $\mathcal{K}^i$ and $\mathcal{T}$ are closed under Minkowski combinations $c_\lambda((K, B, D), (K', B', D')) = ((1 - \lambda)K + \lambda K', \dots)$. A functional on such a set is *convex-linear* if it commutes with c_λ , and *convex*

or *concave* if it satisfies the corresponding inequality for c_λ . The *canonical triple* of $K \in \mathcal{K}^i$ is $\xi_K = (K, B_K, D_K)$. For a convex body K , $a < b < a + \pi$, and a continuous curve $\mathbf{z} : [a, b] \rightarrow \mathbb{R}^2$ of bounded variation with $\mathbf{z}(t) \in l_K(t)$, put

$$\varrho_{K,\mathbf{z}}(t) = \langle \mathbf{z}(t) - v_K^+(t), v_t \rangle, \quad \mathcal{M}_K(a, b; \mathbf{z}) = \frac{1}{2} \int_a^b \varrho_{K,\mathbf{z}}(t)^2 dt.$$

By Mamikon's theorem [1, Thm. 7.4.1] (see [7]), $\mathcal{M}_K(a, b; \mathbf{z})$ is the area swept by the tangent segments from $v_K^+(t)$ to $\mathbf{z}(t)$; we use it only through the formula, which makes $\mathcal{M}_K(a, b; \mathbf{z})$ a convex function of $\varrho_{K,\mathbf{z}}$. The curves that occur are the outer corner $\mathbf{y}_K$ and the tangent line parametrizations $\mathbf{l}_K^T(t) = v_K(t, T)$ for $T - \pi < t < T$ (the point where $l_K(t)$ meets $l_K(T)$) and $\mathbf{l}_K^T(T) = v_K^-(T)$. Let $\mathcal{S}_K, \mathcal{R}_B, \mathcal{L}_D$ be the *Mamikon terms*

$$\begin{aligned} \mathcal{S}_K &= \mathcal{M}_K(0, \varphi^R; \mathbf{l}_K^{\pi/2}) + \mathcal{M}_K(\varphi^R, \varphi^L; \mathbf{y}_K) \\ &\quad + \mathcal{M}_K(\varphi^L, \frac{\pi}{2}; \mathbf{l}_K^{\pi-\varphi}) + \mathcal{M}_K(\frac{\pi}{2}, \pi; \mathbf{l}_K^\pi), \\ \mathcal{R}_B &= \mathcal{M}_B(\pi + \varphi^R, \frac{3\pi}{2}; \mathbf{l}_B^{3\pi/2}), \quad \mathcal{L}_D = \mathcal{M}_D(\frac{3\pi}{2}, \frac{3\pi}{2} + \varphi^L; \mathbf{l}_D^{3\pi/2+\varphi^L}). \end{aligned}$$

(In the third term $\pi/2 + \varphi^L = \pi - \varphi$.) The printed $\mathcal{R}_B$ has $\pi/2 + \varphi^R$ in place of $\pi + \varphi^R$ (E27).

Fact 2.15 (the upper bound). *There is a functional $\mathcal{Q}$ on $\mathcal{T}$ such that:*

- (a) $\xi_K \in \mathcal{T}$ for $K \in \mathcal{K}^i$, and $\mathcal{A}(K) \leq \mathcal{Q}(\xi_K)$. [1, Thm. 8.1.8, Thm. 8.2.4]
- (b) $\mathcal{Q}(K, B, D) = \Lambda(K) - \mathcal{S}_K - \mathcal{R}_B - \mathcal{L}_D$, where Λ (Baek's $\mathcal{P}_K + \mathcal{S}_K$, [1, Def. 8.3.4]) is convex-linear on $\mathcal{K}^i$. For each of the four terms of $\mathcal{S}_K$ (resp. for $\mathcal{R}_B$, for $\mathcal{L}_D$) the curve $\mathbf{z}$ above is convex-linear in K (resp. in B , in D), so that $\varrho(t)$ is convex-linear for every t , and each of the terms $\mathcal{M}$, being $\frac{1}{2} \int \varrho^2$, is convex. [1, Lemma 8.3.3, Lemma 8.3.4, Lemma 8.3.7, Thm. 8.3.1, Thm. 8.3.2, Thm. 7.1.2, Thm. 7.4.1]
- (c) $\mathcal{Q} \leq \mathcal{Q}(\xi_G) = |G|$ on $\mathcal{T}$, where $\xi_G = \xi_{K_G}$ is the canonical triple of Gerver's cap. [1, Cor. 8.5.8, Thm. 8.4.6]

Part (a) is stated in [1] with a proof that uses $\mathcal{N}(K) \subset K$, which $\mathcal{K}^i$ does not provide; the statement is true and has a proof that avoids it, which is in [14] (E21). Also [1, Thm. 8.1.1] (convexity of $\mathcal{K}^i$) uses the Brunn–Minkowski inequality, and it holds as stated.

2.6. Gerver's sofa. Seen from a sofa that turns clockwise by $t \in [0, \pi/2]$, the hallway occupies $\mathbf{x}(t) + R_t L$; the curve $\mathbf{x} : [0, \pi/2] \rightarrow \mathbb{R}^2$ is the *rotation path*. The largest set that moves along $\mathbf{x}$ is its *shape*

$$\text{shape}(\mathbf{x}) = H_L \cap \bigcap_{t \in [0, \pi/2]} (\mathbf{x}(t) + R_t L) \cap (\mathbf{x}(\pi/2) + R_{\pi/2} V_L).$$

Gerver's sofa G is the shape of Romik's rotation path [10], which is glued from five explicit curves, on $[0, \varphi]$, $[\varphi, \theta]$, $[\theta, \pi/2 - \theta]$, $[\pi/2 - \theta, \pi/2 - \varphi]$ and $[\pi/2 - \varphi, \pi/2]$, with $\varphi = 0.03917\dots$ and $\theta = 0.68130\dots$ (the formulas and

Romik's equations are in Appendix A). Write $t_0, \dots, t_5 = 0, \varphi, \theta, \pi/2 - \theta, \pi/2 - \varphi, \pi/2$, $\alpha = \langle \mathbf{x}', u_t \rangle$, $\beta = \langle \mathbf{x}', v_t \rangle$, and

$$\mathbf{A} = \mathbf{x} + \alpha v_t + u_t, \quad \mathbf{B} = \mathbf{x} + \alpha v_t, \quad \mathbf{C} = \mathbf{x} - \beta u_t + v_t, \quad \mathbf{D} = \mathbf{x} - \beta u_t$$

for the contact points of the sofa with the walls of the hallway.

Fact 2.16 (Gerver's sofa). *(a) Romik's equations have exactly one solution with $\varphi \in [0.039, 0.04]$, $\theta \in [0.68, 0.69]$, so G is well defined, and $2.2192 \leq |G| \leq 2.2199$. (Romik asserts, without proof, the uniqueness for $0 < \varphi < \theta < \pi/4$; uniqueness in this box is proved in [14].)*

(b) G is a monotone sofa with rotation angle $\pi/2$. Its cap $K_G = \mathcal{C}(G)$ satisfies $G = K_G \setminus \mathcal{N}(K_G)$ and has $A_{K_G} = \mathbf{A}$, $C_{K_G} = \mathbf{C}$ and $\mathbf{x}_{K_G} = \mathbf{x}$ on $[0, \pi/2]$; and K_G satisfies the injectivity condition; as $|K_G| \geq |G| > 11/5$, $K_G \in \mathcal{K}^i$. [1, Thm. 8.4.1 (1), Thm. 6.1.2]

(c) Let $\Gamma = \mathbf{D}([t_0, t_2]) \cup \mathbf{x}([t_1, t_4]) \cup \mathbf{B}([t_3, t_5])$. Then $\mathbf{D}(t_0)$ and $\mathbf{B}(t_5)$ lie on the x -axis, $\mathbf{D}(t_2) = \mathbf{x}(t_4)$, $\mathbf{B}(t_3) = \mathbf{x}(t_1)$, the abscissa is strictly increasing along $\mathbf{D}$ and $\mathbf{B}$ and strictly decreasing along $\mathbf{x}$ on $(0, \pi/2)$, at every point of $[t_0, t_1]$ and of $[t_1, t_2]$ the curve $\mathbf{D}$ has a derivative within that interval that is a positive multiple of u_t , and at every point of $[t_3, t_4]$ and of $[t_4, t_5]$ the curve $\mathbf{B}$ has a derivative within that interval that is a negative multiple of v_t (the derivatives are one-sided at the ends of the pieces, and jump at t_1 and t_4), and

$$\mathcal{N}(K_G) = \{q : q_y \geq 0, q_y < \gamma_y \text{ for some } \gamma \in \Gamma \text{ with } \gamma_x = q_x\}.$$

[1, Thm. 8.4.1 (2)–(4)]

Baek states [1, Thm. 8.4.1] without proof (his Remark 8.4.1: the properties are “easy to verify numerically”), and proves [1, Thm. 6.1.2] by a misreading of a theorem of Gerver (E12, E24). Both statements are true; the formalization proves them from Romik's equations by interval arithmetic [14]. Baek's [1, Thm. 8.4.1 (2)] describes the niche as the region enclosed by $\mathbf{B}$ reversed, $\mathbf{x}|_{[t_1, t_4]}$, $\mathbf{D}$ reversed and the segment of the x -axis from $\mathbf{D}(t_0)$ to $\mathbf{B}(t_5)$. That open region omits the open segment of the x -axis between $\mathbf{D}(t_0)$ and $\mathbf{B}(t_5)$, which lies in $\mathcal{N}(K_G)$ because $F_{\pi/2}$ is closed. Because of the monotonicity of the abscissa, $\mathcal{N}(K_G)$ is the set of points of the closed upper half-plane that lie strictly below the curve Γ , which is the identity in (c); this exact form is proved in [14]. We also use one numerical fact that is not in [1]:

Fact 2.17 (height of the rotation path). $\mathbf{x}(t)_y < 1$ for every $t \in [0, \pi/2]$, and $\mathbf{x}(t)_y > 0$ for $t \in [t_1, t_4]$. In fact $\max_t \mathbf{x}(t)_y = \mathbf{x}(\pi/4)_y = 0.66426\dots$ and $\mathbf{x}(t_1)_y = \mathbf{x}(t_4)_y = 0.05518\dots$, but we use only the two bounds, which the formalization proves by interval arithmetic [14]: $\mathbf{x}(t)_y \leq 0.9925$ on each phase, and $\mathbf{x}_y > 0$ on $[t_1, t_4]$.

3. FROM A MAXIMAL SOFA TO A MAXIMIZING CAP

Throughout this section S is a moving sofa with $|S| = |G|$. We first reduce the problem to the caps, and then outline the rest of the proof.

Definition 3.1 (maximizing cap). Let $\omega \in (0, \pi/2]$. A cap $K \in \mathcal{K}_\omega^c$ is *maximizing* if $\mathcal{A}_\omega(C) \leq \mathcal{A}_\omega(K)$ for every $C \in \mathcal{K}_\omega^c$.

By Fact 2.7, every cap with $\mathcal{A}_\omega(K) = |G|$ is maximizing. Note that a maximizing cap is a maximizer of $\mathcal{A}_\omega$ over *all* caps, not only over those that contain their niche. Baek constructs a particular maximizing cap, a limit of maximum polygon caps; a given maximizing cap need not be such a limit, and this is the point where the argument of [1] does not give uniqueness.

Proposition 3.2 (the monotonization of a maximal sofa). *Let S' be a moving sofa with rotation angle $\omega \in (0, \pi/2]$ and $|S'| = |G|$. Then there are $w \in \mathbb{R}^2$ and a monotone sofa T with rotation angle ω such that*

$$S' + w \subset T, \quad |T| = |G|,$$

and the cap $K = \mathcal{C}(T)$ is a maximizing cap, with $T = K \setminus \mathcal{N}(K)$ and $\mathcal{A}_\omega(K) = |G|$. In particular, if S is any moving sofa with $|S| = |G|$, there is such a T for $S' = S$ and some $\omega \in [\omega_0, \pi/2]$.

Proof. By Fact 2.3(b), $S' + w$ is in standard position for some w , and is a moving sofa with rotation angle ω . By Fact 2.4(a), $T = \mathcal{I}(S' + w)$ is a moving sofa in standard position containing $S' + w$, and it is a monotone sofa. So $|G| = |S' + w| \leq |T| \leq |G|$ by Fact 2.7. By Fact 2.4(b), $K = \mathcal{C}(T)$ is a cap, and Fact 2.4(c) gives $T = K \setminus \mathcal{N}(K)$ and $\mathcal{A}_\omega(K) = |T| = |G|$; K is maximizing by Fact 2.7. Finally, if $|S| = |G|$ then $|S| > 11/5$, so S has a rotation angle $\omega \in [\omega_0, \pi/2]$ by Fact 2.3(a). $\square$

So the problem is to show that S is a rigid image of G . The proof follows S through a chain of sets, each of which contains a rigid image of S and has area $|G|$ (Figure 3). Steps (1) and (2) are used twice: first for the maximizing cap K of Proposition 3.2, which has the rotation angle ω , to obtain step (3), and then, after step (3), for the maximizing cap of the new monotone sofa U , which has the rotation angle $\pi/2$, to obtain step (4).

- (1) *Selection* (Section 4). A given maximizing cap K_* with $\mathcal{A}_\omega(K_*) > 0$ is the Hausdorff limit of polygon caps K_n that maximize the polygon sofa area minus a penalty for straying from K_* .
- (2) *Variations* (Section 5). Pushing one side of K_n shows that K_n is almost balanced: the defect $\sigma_{K_n} - \tau_{K_n}$ at a normal is bounded by a multiple of the distance from K_n to K_* at the sampled normals, and the weights are chosen so that these bounds add up.
- (3) *The right angle* (Section 6). If $\omega < \pi/2$, the limit of the defect bounds gives the two *pinned bounds* $w_{K_*}^\circ \leq \sigma_{K_*}(\pi/2)$ and $z_{K_*}^\circ \leq \sigma_{K_*}(\omega)$ between wedge gaps and side lengths of K_* . They give T width at most one in the directions u_t , $t \in [\omega, \pi/2]$, so that a rotated copy $R_\psi T$, $\psi = \pi/2 - \omega$, turns through $\pi/2$. Monotonizing $R_\psi T$ gives a monotone sofa U with rotation angle $\pi/2$ and $|U| = |G|$ that contains a rigid image of S .

- (4) *Injectivity* (Section 7). For the cap K of U , the defect bounds become an inequality $\sigma_K \leq k_0(g_K^+(t))dt$ on $[0, \pi/2)$, and Baek's iteration gives the injectivity condition.
- (5) *Equality* (Section 8). Then $|G| = \mathcal{A}(K) \leq \mathcal{Q}(\xi_K) \leq \mathcal{Q}(\xi_G) = |G|$, and equality in the concave functional $\mathcal{Q}$ forces equality in each Mamikon term, a convex integral $\frac{1}{2} \int \varrho^2$. This gives differential equations for the difference of the support functions of K and of Gerver's cap, whose solutions make K a horizontal translate of Gerver's cap and U a horizontal translate of G .
- (6) *Recovery* (Section 9). A closed subset of G with the area of G is G , since G is the closure of its interior.

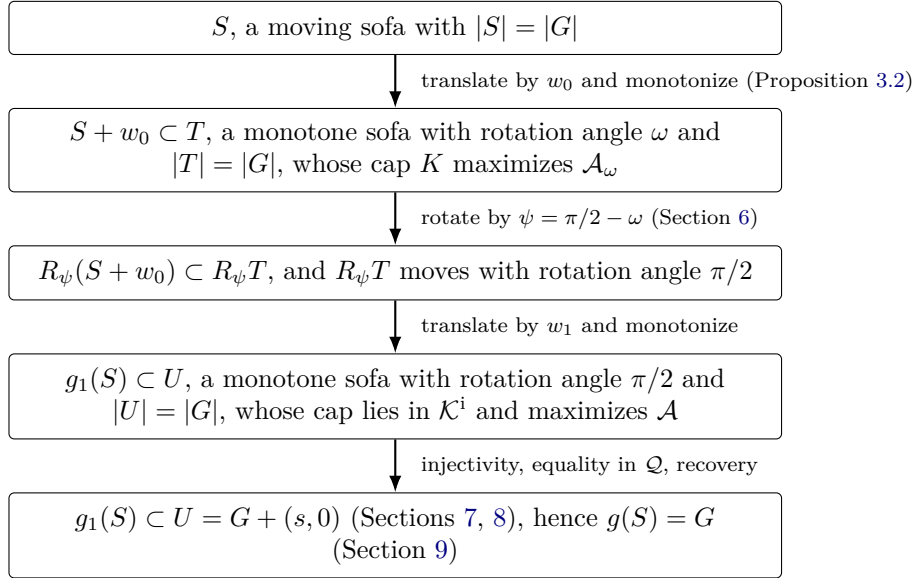


FIGURE 3. The containment chain of the proof. Every set contains the image of S under the maps so far, and all the sets have area $|G|$; the first three arrows are a translation, a rotation or a monotone. Here $g_1(p) = R_\psi(p + w_0) + w_1$ and $g(p) = g_1(p) - (s, 0)$.

4. APPROXIMATING A MAXIMIZING CAP BY POLYGON CAPS

In this section and in Section 5 we fix $\omega \in (0, \pi/2]$ and a maximizing cap $K_* \in \mathcal{K}_\omega^c$ with $\mathcal{A}_\omega(K_*) > 0$, and write $h_* = h_{K_*}$. Section 6 uses them for the cap of Proposition 3.2, which qualifies since $|G| > 11/5$, and Section 7 uses them again, with $\omega = \pi/2$, for the cap of the monotone sofa U of Proposition 6.9, which qualifies since its sofa area is $|G| > 0$.

Baek approximates a cap by maximum polygon caps and proves that these are balanced; he then takes the limit of a particular sequence of maximum

polygon caps. A maximizing cap that is not such a limit is not reached this way: if $F_n \downarrow F$ uniformly on $[0, 1]$, a maximizer of F need not be a limit of maximizers of the F_n (take $F_n(x) = x/n$ and $F = 0$, whose maximizers are all of $[0, 1]$ while those of F_n are $\{1\}$). We repair this with a penalty that vanishes at the chosen maximizer, as in the example $F_n(x) - (x - x_*)^2$, whose maximizers converge to x_* . The penalty is a weighted sum of squared differences of support values at dyadic normals; Section 5 shows how it spoils the balance.

4.1. Dyadic angles and the penalty. For $k \geq 0$ let $\Theta_k = \{i\omega/2^{k+1} : 0 < i < 2^{k+1}\}$, an angle set with 2^{k+1} intervals and $|\Theta_k| = 2^{k+1} - 1$. Then $\Theta_k \subset \Theta_{k+1}$, $\omega/2 \in \Theta_k$, and $\bigcup_k \Theta_k$ is dense in $[0, \omega]$. Write $\mathcal{K}_k^c = \mathcal{K}_{\Theta_k}^c$ for the polygon caps with angle set Θ_k and Θ_k° for the defining normals.

For $m \geq 0$ let $\nu_m = 2^{-m-1}/(2(2^{m+1} - 1))$. For a convex body K and $k \geq 0$ let

$$\Pi_k(K) = \sum_{m=0}^k \nu_m \sum_{t \in \Theta_m} \left[(h_K(t) - h_*(t))^2 + (h_K(t + \frac{\pi}{2}) - h_*(t + \frac{\pi}{2}))^2 \right].$$

The terms are the *samples*; an angle of level m also lies in every later level, so it carries one sample from each level between its first and k . For $s \in \mathbb{R}$ let $\nu_k(s) \geq 0$ be the total weight of the samples taken at the normal s , so that $\nu_k(s) = 0$ unless $s \in \Theta_k \cup (\Theta_k + \pi/2)$. The samples of level m have total weight $2|\Theta_m|\nu_m = 2^{-m-1}$, hence

$$\sum_s \nu_k(s) = \sum_{m=0}^k 2^{-m-1} = 1 - 2^{-k-1} < 1. \quad (4.1)$$

Two elementary facts about Π_k will be used. First, *persistence*: if $m \leq k$ and $t \in \Theta_m$, then

$$\nu_m(h_K(s) - h_*(s))^2 \leq \Pi_k(K) \quad (s = t \text{ and } s = t + \pi/2), \quad (4.2)$$

because every term of Π_k is nonnegative. Second, if $|h_K(s) - h_*(s)| \leq \eta$ at every sample, and K' is a convex body, then

$$|\Pi_k(K') - \Pi_k(K)| \leq \sum_s \nu_k(s) (2\eta r_s + r_s^2), \quad r_s = |h_{K'}(s) - h_K(s)|, \quad (4.3)$$

where the sum is over the sampled normals s ; this is the identity $(b-c)^2 - (a-c)^2 = 2(a-c)(b-a) + (b-a)^2$ applied to each term. In particular, if K' has the same support values as K at every sampled normal other than one normal s_0 , where they differ by at most r , then $|\Pi_k(K') - \Pi_k(K)| \leq \nu_k(s_0)(2\eta r + r^2)$.

Definition 4.1 (penalized maximizer). A polygon cap $K \in \mathcal{K}_k^c$ is a *penalized maximizer at stage k* if $\mathcal{A}_{\Theta_k}(C) - \Pi_k(C) \leq \mathcal{A}_{\Theta_k}(K) - \Pi_k(K)$ for every $C \in \mathcal{K}_k^c$.

Lemma 4.2 (recovery polygon). $K_k^r = \mathcal{C}_{\Theta_k}(K_*) \in \mathcal{K}_k^c$ has $\Pi_k(K_k^r) = 0$ and $\mathcal{A}_{\Theta_k}(K_k^r) \geq \mathcal{A}_\omega(K_*)$.

Proof. By Fact 2.8, K_k^r is a polygon cap with the support values of K_* on $\Theta_k^\diamond$, which contain all the sampled normals; so $\Pi_k(K_k^r) = 0$. Also $\mathcal{A}_{\Theta_k}(K_k^r) = \mathcal{A}_{\Theta_k}(K_*) \geq \mathcal{A}_\omega(K_*)$. $\square$

Lemma 4.3 (two samples bound a cap). *Let $K \in \mathcal{K}_k^c$, $t \in \Theta_k$, and suppose that $q(h_K(s) - h_*(s))^2 \leq c$ for $s = t$ and $s = t + \pi/2$, with $q > 0$, $c \geq 0$. Then $K \subset [-R, R] \times [0, 1]$ with $R = H/\sin t + H/\cos t$, $H = |h_*(t)| + |h_*(t + \pi/2)| + c/q + 1$.*

Proof. From $x^2 \leq c/q$ follows $|x| \leq c/q + 1$, so $h_K(t) \leq H$ and $h_K(t + \pi/2) \leq H$. A point $(x, y) \in K$ has $0 \leq y \leq 1$, since $K \subset P_\omega$. As $t \in (0, \pi/2)$, $x \cos t + y \sin t \leq H$ gives $x \leq H/\cos t$, and $-x \sin t + y \cos t \leq H$ gives $-x \leq H/\sin t$. $\square$

Lemma 4.4 (the niche lies in a box). *If $K \in \mathcal{K}_\omega^c$ and $K \subset [-R, R] \times [0, 1]$, then $\mathcal{N}(K) \subset [-R, R] \times [0, R]$; the same holds for $\mathcal{N}_\Theta(K) \subset \mathcal{N}(K)$.*

Proof. Let $p \in \mathcal{N}(K)$, so $p_y \geq 0$ and $p \in Q_K^-(t)$ for some $t \in (0, \omega)$. As $K \subset [-R, R] \times [0, 1]$, $h_K(t) \leq R \cos t + \sin t \leq R \cos t + 1$ and $h_K(t + \pi/2) \leq R \sin t + \cos t \leq R \sin t + 1$. So $\langle p, u_t \rangle < R \cos t$ and $\langle p, v_t \rangle < R \sin t$. Since $p_y \geq 0$ and $\langle p, u_t \rangle = p_x \cos t + p_y \sin t$, we get $p_x < R$; likewise $\langle p, v_t \rangle = -p_x \sin t + p_y \cos t$ gives $p_x > -R$. Finally $p_y = \sin t \langle p, u_t \rangle + \cos t \langle p, v_t \rangle < 2R \sin t \cos t \leq R$. $\square$

The proof of the next proposition uses the bounded width of polygon caps of positive area (Fact 2.9), whose proof needs the fan in the definition of $\mathcal{N}_\Theta$ (E6).

Proposition 4.5 (penalized maximizers exist). *For every $k \geq 0$ there is a penalized maximizer $K^{(k)}$ at stage k , and*

$$\mathcal{A}_{\Theta_k}(K^{(k)}) - \Pi_k(K^{(k)}) \geq \mathcal{A}_\omega(K_*).$$

There is $R_0 < \infty$ such that $K^{(k)} \subset [-R_0, R_0] \times [0, 1]$ for every k .

Proof. Let $\Psi_k = \mathcal{A}_{\Theta_k} - \Pi_k$ on $\mathcal{K}_k^c$, and $\bar{t} = \omega/2 \in \Theta_k$; the constant $c = c_{\omega, \bar{t}}$ of Fact 2.9 does not depend on k . Let $C \in \mathcal{K}_k^c$ with $\Psi_k(C) > 0$. Then $\mathcal{A}_{\Theta_k}(C) > 0$, so C has width at most c and, as it lies in a strip of height one, $|C| \leq c$; hence $\Pi_k(C) < \mathcal{A}_{\Theta_k}(C) \leq |C| \leq c$ and $\Psi_k(C) \leq c$. The samples of level 0 lie at $\bar{t}$ and $\bar{t} + \pi/2$ and have weight $\nu_0 = 1/4$ at every stage, so by (4.2) $\nu_0(h_C(s) - h_*(s))^2 \leq \Pi_k(C) \leq c$ for $s = \bar{t}, \bar{t} + \pi/2$, and Lemma 4.3 puts C in $[-R_0, R_0] \times [0, 1]$ with R_0 independent of k and C .

By Lemma 4.2, $M_k = \sup \Psi_k \geq \Psi_k(K_k^r) \geq \mathcal{A}_\omega(K_*) > 0$, and $M_k \leq c$. Take $C_j \in \mathcal{K}_k^c$ with $\Psi_k(C_j) > 0$ and $\Psi_k(C_j) \rightarrow M_k$. They lie in the box, so by Blaschke's theorem a subsequence converges in d_H to a convex body K_∞ . We show that K_∞ is a penalized maximizer. First, K_∞ is a polygon cap with angle set Θ_k . Its support values at $\omega, \pi/2, \omega + \pi, 3\pi/2$ are 1, 1, 0, 0 as those of the C_j , and $|K_\infty| = \lim |C_j| \geq \lim \Psi_k(C_j) = M_k > 0$ since $|C_j| \geq \mathcal{A}_{\Theta_k}(C_j) \geq \Psi_k(C_j)$, so K_∞ has an interior point q . Let p lie in $K'_\infty = \bigcap_s H_{K_\infty}(s)$, the intersection over $s \in \Theta_k^\diamond \cup \{\omega + \pi, 3\pi/2\}$. For $0 \leq \lambda < 1$ the point $q + \lambda(p - q)$ satisfies $\langle \cdot, u_s \rangle < h_{K_\infty}(s)$ for all these s , hence

$\langle \cdot, u_s \rangle \leq h_{C_j}(s)$ for large j , so it lies in C_j (a polygon cap is the intersection of its supporting half-planes at these normals), hence in K_∞ . As K_∞ is closed, $p \in K_\infty$; so $K_\infty = K'_\infty$ is a polygon cap. Second, $\Psi_k(K_\infty) \geq \limsup \Psi_k(C_j)$. The penalty is continuous in the support values. The area is continuous. Each point of $\mathcal{N}_{\Theta_k}(K_\infty)$ lies in $\mathcal{N}_{\Theta_k}(C_j)$ for all large j : the two strict inequalities defining $Q_{K_\infty}^-(t)$ hold at the point, and persist when the support values converge. These niches lie in $[-R_0, R_0] \times [0, R_0]$ by Lemma 4.4 (so their areas are finite), and Fatou's lemma gives $|\mathcal{N}_{\Theta_k}(K_\infty)| \leq \liminf |\mathcal{N}_{\Theta_k}(C_j)|$. Hence $\mathcal{A}_{\Theta_k}(K_\infty) \geq \limsup \mathcal{A}_{\Theta_k}(C_j)$ and $\Psi_k(K_\infty) \geq M_k$. Put $K^{(k)} = K_\infty$. $\square$

4.2. Convergence to the given cap.

Lemma 4.6 (limits of polygon caps with nested angle sets). *Let $k_1 < k_2 < \dots$ and let $C_j \in \mathcal{K}_{k_j}^c$ lie in a common box $[-R, R] \times [0, 1]$ and converge in d_H to $K_\infty \in \mathcal{K}_\omega^c$. Then $\limsup_j \mathcal{A}_{\Theta_{k_j}}(C_j) \leq \mathcal{A}_\omega(K_\infty)$.*

Proof. By continuity of the area, $|C_j| \rightarrow |K_\infty|$. Let $p \in \mathcal{N}(K_\infty)$: $p \in F_\omega \cap Q_{K_\infty}^-(t)$ for some $t \in (0, \omega)$. The two strict inequalities that define $Q_{K_\infty}^-(t)$ hold at p with a margin $\delta > 0$ at some dyadic angle t' near t , since h_{K_∞} is continuous. For large j , $t' \in \Theta_{k_j}$ and $\sup |h_{C_j} - h_{K_\infty}| < \delta/2$, so $p \in F_\omega \cap Q_{C_j}^-(t') \subset \mathcal{N}_{\Theta_{k_j}}(C_j)$. The sets $\mathcal{N}_{\Theta_{k_j}}(C_j)$ lie in $[-R, R] \times [0, R]$ by Lemma 4.4 (so their areas are finite), and Fatou's lemma gives $|\mathcal{N}(K_\infty)| \leq \liminf |\mathcal{N}_{\Theta_{k_j}}(C_j)|$. Subtract. $\square$

Lemma 4.7 (dyadic supports determine a cap). *If $K, K' \in \mathcal{K}_\omega^c$ and $h_K(s) = h_{K'}(s)$ for all $s \in \bigcup_m (\Theta_m \cup (\Theta_m + \pi/2))$, then $K = K'$.*

Proof. By continuity and density, $h_K = h_{K'}$ on J_ω , and $h_K = h_{K'} = 0$ at $\omega + \pi$ and $3\pi/2$. A cap is the intersection of its supporting half-planes at the normals of $J_\omega \cup \{\omega + \pi, 3\pi/2\}$. $\square$

Proposition 4.8 (selection). *There are $k_1 < k_2 < \dots$ and penalized maximizers K_j at stage k_j , all in $[-R_0, R_0] \times [0, 1]$, such that $K_j \rightarrow K_*$ in d_H .*

Proof. Take the penalized maximizers $K^{(k)}$ of Proposition 4.5. By Blaschke's theorem there are $k_1 < k_2 < \dots$ with $K_j = K^{(k_j)} \rightarrow K_\infty$ for a convex body K_∞ .

K_∞ is a cap with rotation angle ω . The support values at $\omega, \pi/2, \omega + \pi, 3\pi/2$ pass to the limit. Let (a, b) be a gap of $J_\omega \cup \{\omega + \pi, 3\pi/2\}$ in the circle; it has length at most $\pi/2$. A polygon cap K_j has no normal angle strictly between a and b , so the supporting lines $l_{K_j}(a)$ and $l_{K_j}(b)$ meet at a vertex q of K_j and $h_{K_j}(r) = \langle q, u_r \rangle$ for $a \leq r \leq b$; with $\sin(b-a)u_r = \sin(b-r)u_a + \sin(r-a)u_b$ this gives $\sin(b-a)h_{K_j}(r) = \sin(b-r)h_{K_j}(a) + \sin(r-a)h_{K_j}(b)$. The identity passes to h_{K_∞} , and its coefficients are nonnegative, so $H_{K_\infty}(a) \cap H_{K_\infty}(b) \subset H_{K_\infty}(r)$. Hence $K_\infty = \bigcap_r H_{K_\infty}(r)$ is the intersection of the $H_{K_\infty}(r)$ with $r \in J_\omega \cup \{\omega + \pi, 3\pi/2\}$.

The penalties tend to 0. By Proposition 4.5, $0 \leq \Pi_{k_j}(K_j) \leq \mathcal{A}_{\Theta_{k_j}}(K_j) - \mathcal{A}_\omega(K_*)$. By Lemma 4.6 and the maximality of K_* , $\limsup_j \mathcal{A}_{\Theta_{k_j}}(K_j) \leq \mathcal{A}_\omega(K_\infty) \leq \mathcal{A}_\omega(K_*)$. So $\Pi_{k_j}(K_j) \rightarrow 0$.

$K_\infty = K_*$. Let $t \in \Theta_m$. For $k_j \geq m$, (4.2) gives $\nu_m(h_{K_j}(s) - h_*(s))^2 \leq \Pi_{k_j}(K_j) \rightarrow 0$ for $s = t, t + \pi/2$, while $h_{K_j}(s) \rightarrow h_{K_\infty}(s)$. So $h_{K_\infty} = h_*$ at all dyadic normals, and $K_\infty = K_*$ by Lemma 4.7. The box is Proposition 4.5. $\square$

Put $\eta_j = d_H(K_j, K_*) \rightarrow 0$. At every sampled normal, $|h_{K_j}(s) - h_*(s)| \leq \eta_j$.

5. PENALIZED MAXIMIZERS ARE ALMOST BALANCED

A maximum polygon cap is balanced [1, Thm. 3.4.9]: $\sigma_K(t) = \tau_K(t)$ for every $t \in \Theta^\circ$, since otherwise pushing one side would increase $\mathcal{A}_\Theta$. A penalized maximizer need not be balanced, but pushing a side changes the penalty by a controlled amount, and the argument gives a one-sided bound on the defect $\zeta_K(t) = \sigma_K(t) - \tau_K(t)$. In this section we fix $k \geq 0$, $\Theta = \Theta_k$, a penalized maximizer $K \in \mathcal{K}_k^c$ at stage k , and $\eta \geq 0$ with $|h_K(s) - h_*(s)| \leq \eta$ at every sampled normal s . Let $\bar{\nu} = \sum_s \nu_k(s) < 1$ be the total weight, (4.1).

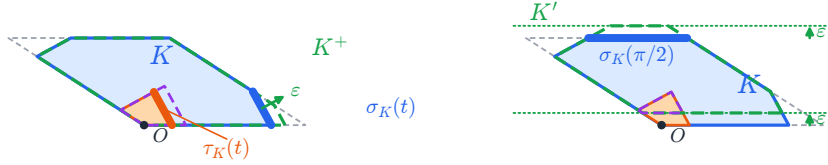


FIGURE 4. Pushing one side of a polygon cap K with angle set $\{\omega/2\}$, $\omega = 1$ (the dashed parallelogram is P_ω ; the niche is orange). Left: raising the height at the floating normal $t = \omega/2$ by ε gives the cap K^+ (dashed); it gains a strip of area $\sigma_K(t)\varepsilon + O(\varepsilon^2)$ along its edge with normal t , and its niche a strip of area $\tau_K(t)\varepsilon + O(\varepsilon^2)$ along the side on the inner wall $b_K(t)$. Right: raising the pinned height at $\pi/2$ moves both lines of the horizontal strip up by ε ; the cap gains a strip along its top edge, of length $\sigma_K(\pi/2)$, and loses one along its bottom edge, and the result is a translate of a polygon cap.

5.1. Floating normals.

Lemma 5.1 (raising a floating height). *Let $t \in \Theta^\circ \setminus \{\omega, \pi/2\}$, $\varepsilon > 0$, and $K^+ = \mathcal{C}_\Theta(h_K^{t,\varepsilon})$. Then $K^+ \in \mathcal{K}_\Theta^c$, $K \subset K^+$, $h_{K^+}(s) = h_K(s)$ for $s \in \Theta^\circ \setminus \{t\}$, and $h_K(t) \leq h_{K^+}(t) \leq h_K(t) + \varepsilon$.*

Proof. Since $h_K(\omega) = h_K(\pi/2) = 1$, $P_{h_K^{t,\varepsilon}} = P_\omega$, so

$$K^+ = P_\omega \cap \bigcap_s H_-(s, h_K^{t,\varepsilon}(s)),$$

the intersection over the floating normals s . Then $K = \mathcal{C}_\Theta(h_K) \subset K^+ \subset P_\omega$. So K^+ is a convex body (it is bounded, since the half-planes with normals t' and $t' + \pi/2$, $t' \in \Theta$, bound it also when $\omega = \pi/2$), it has the support values 1, 1, 0, 0 of P_ω at $\omega, \pi/2, \omega + \pi, 3\pi/2$, and it is an intersection of half-planes with the normal angles of a polygon cap. So $K^+ \in \mathcal{K}_\Theta^c$. Next, $h_{K^+} \geq h_K$ since $K \subset K^+$, and $h_{K^+}(s) \leq h_K^{t,\varepsilon}(s)$ since $K^+ \subset H_-(s, h_K^{t,\varepsilon}(s))$. $\square$

Proposition 5.2 (floating defects). *For every floating normal $t \in \Theta^\diamond \setminus \{\omega, \pi/2\}$,*

$$\sigma_K(t) - \tau_K(t) \leq 2\eta \nu_k(t).$$

Proof. Let $K_\varepsilon^+ = \mathcal{C}_\Theta(h_K^{t,\varepsilon})$. By Lemma 5.1, K_ε^+ is a polygon cap that has the support values of K on $\Theta^\diamond \setminus \{t\}$, and differs from K at t by at most ε ; so by (4.3), $\Pi_k(K_\varepsilon^+) - \Pi_k(K) \leq \nu_k(t)(2\eta\varepsilon + \varepsilon^2)$. Penalized maximality gives $\mathcal{A}_\Theta(K_\varepsilon^+) - \mathcal{A}_\Theta(K) \leq \Pi_k(K_\varepsilon^+) - \Pi_k(K)$. By Fact 2.11(c) (with $C = K_\varepsilon^+$, $v = 0$) and (a), for small ε

$$\mathcal{A}_\Theta(K_\varepsilon^+) \geq \mathcal{A}_\Theta(h_K^{t,\varepsilon}) \geq \mathcal{A}_\Theta(K) + (\sigma_K(t) - \tau_K(t))\varepsilon - M\varepsilon^2.$$

Hence $(\sigma_K(t) - \tau_K(t))\varepsilon - M\varepsilon^2 \leq \nu_k(t)(2\eta\varepsilon + \varepsilon^2)$. Divide by ε and let $\varepsilon \rightarrow 0$. $\square$

5.2. Pinned normals. In this subsection $\omega < \pi/2$, so that the two pinned normals ω and $\pi/2$ are distinct and every cap contains O and o_ω (Lemma 2.5); in particular $h_K \geq 0$. Moving a pinned normal moves both lines of a strip, and the result is in general not a polygon cap but a translate of one.

Lemma 5.3 (the pinned sandwich). *Let $K \in \mathcal{K}_\Theta^c$, $t \in \{\omega, \pi/2\}$, $\varepsilon \in [0, 1]$, and $K' = \mathcal{C}_\Theta(h_K^{t,\varepsilon})$. Then*

$$(1 - \varepsilon)K + \varepsilon o_\omega \subset K' \subset (1 + \varepsilon)K.$$

If $K \subset \{\|p\| \leq R\}$, then $|h_{K'}(s) - h_K(s)| \leq 2R\varepsilon$ for every s .

Proof. Let t' be the other pinned normal. Then $K' = \{p : \varepsilon \leq \langle p, u_t \rangle \leq 1 + \varepsilon, 0 \leq \langle p, u_{t'} \rangle \leq 1, \langle p, u_s \rangle \leq h_K(s) \text{ for floating } s\}$. Let $p \in K$ and $q = (1 - \varepsilon)p + \varepsilon o_\omega \in K'$. Then q satisfies every inequality of K' except possibly the one for t , and $\langle q, u_t \rangle = (1 - \varepsilon)\langle p, u_t \rangle + \varepsilon \in [\varepsilon, 1]$ because $\langle o_\omega, u_t \rangle = 1$ and $0 \leq \langle p, u_t \rangle \leq 1$. So $q \in K'$. Conversely, let $p \in K'$. For floating s , $\langle p/(1 + \varepsilon), u_s \rangle \leq h_K(s)/(1 + \varepsilon) \leq h_K(s)$ as $h_K(s) \geq 0$; $\langle p, u_t \rangle/(1 + \varepsilon) \in [0, 1]$ and $\langle p, u_{t'} \rangle/(1 + \varepsilon) \in [0, 1]$. So $p/(1 + \varepsilon) \in K$. Taking support functions, $(1 - \varepsilon)h_K(s) + \varepsilon \langle o_\omega, u_s \rangle \leq h_{K'}(s) \leq (1 + \varepsilon)h_K(s)$, and $|\langle o_\omega, u_s \rangle| \leq R$ since $o_\omega \in K$. $\square$

Lemma 5.4 (moving back to standard position). *In Lemma 5.3, suppose $K' = C + v$ with $C \in \mathcal{K}_\Theta^c$ and $v \in \mathbb{R}^2$. Then $|\langle v, u_s \rangle| \leq (2/\cos \omega + 1)\varepsilon$ for all s , and if $K \subset \{\|p\| \leq R\}$,*

$$|h_C(s) - h_K(s)| \leq \gamma_\omega \varepsilon \quad \text{for every } s, \quad \gamma_\omega = 2R + \frac{2}{\cos \omega} + 1.$$

Proof. For $s \in \{\omega, \pi/2\}$, K' lies in the strip $a_s \leq \langle p, u_s \rangle \leq a_s + 1$, where $a_t = \varepsilon$ and $a_{t'} = 0$. As C is a cap, it has width exactly 1 in the direction u_s (its support values at s and $s+\pi$ are 1 and 0), and so has $C+v$; hence $\langle v, u_s \rangle = a_s$. With $s = \pi/2$ and $s = \omega$ this gives $v_y \in [0, \varepsilon]$ and $v_x \cos \omega + v_y \sin \omega \in [0, \varepsilon]$, so $|v_x| \leq 2\varepsilon / \cos \omega$ and $|\langle v, u_s \rangle| \leq |v_x| + |v_y|$. Finally $h_C(s) = h_{K'}(s) - \langle v, u_s \rangle$, and Lemma 5.3 bounds $h_{K'}(s) - h_K(s)$. $\square$

Proposition 5.5 (pinned defects). *Let $\omega < \pi/2$, $K \subset \{\|p\| \leq R\}$, and $t \in \{\omega, \pi/2\}$. Then*

$$\sigma_K(t) - \tau_K(t) \leq 2\bar{\nu}\eta\gamma_\omega, \quad \gamma_\omega = 2R + \frac{2}{\cos \omega} + 1.$$

Proof. If $\sigma_K(t) = 0$, the left side is $-\tau_K(t) \leq 0$. Let $\sigma_K(t) > 0$. By Fact 2.11(b), for small $\varepsilon > 0$ the set $\mathcal{C}_\Theta(h_K^{t,\varepsilon})$ is a translate $C_\varepsilon + v_\varepsilon$ of a polygon cap C_ε . By Lemma 5.4, $|h_{C_\varepsilon}(s) - h_K(s)| \leq \gamma_\omega \varepsilon$ at every s , so (4.3) gives $\Pi_k(C_\varepsilon) - \Pi_k(K) \leq \bar{\nu}(2\eta\gamma_\omega \varepsilon + \gamma_\omega^2 \varepsilon^2)$. Penalized maximality compares K with C_ε , and Fact 2.11(c), (a) give

$$\mathcal{A}_\Theta(C_\varepsilon) \geq \mathcal{A}_\Theta(h_K^{t,\varepsilon}) \geq \mathcal{A}_\Theta(K) + (\sigma_K(t) - \tau_K(t))\varepsilon - M\varepsilon^2.$$

So $(\sigma_K(t) - \tau_K(t))\varepsilon - M\varepsilon^2 \leq \bar{\nu}(2\eta\gamma_\omega \varepsilon + \gamma_\omega^2 \varepsilon^2)$; divide by ε and let $\varepsilon \rightarrow 0$. $\square$

5.3. A two-sided bound.

Corollary 5.6 (two-sided defect bound). *Let $\omega < \pi/2$ and $K \subset \{\|p\| \leq R\}$. For every $t \in \Theta^\circ$,*

$$|\sigma_K(t) - \tau_K(t)| \leq \frac{2\bar{\nu}\eta + 4\bar{\nu}\eta\gamma_\omega}{\sin t}.$$

Proof. Put $\zeta(s) = \sigma_K(s) - \tau_K(s)$, and $e(s) = 2\eta\nu_k(s)$ for floating s , $e(s) = 2\bar{\nu}\eta\gamma_\omega$ for the two pinned s . By Propositions 5.2 and 5.5, $\zeta \leq e$ on Θ° , and $e \geq 0$ with $\sum_s e(s) \leq 2\bar{\nu}\eta + 4\bar{\nu}\eta\gamma_\omega =: E$. By Fact 2.10(c), $\sum_s \sin s \zeta(s) = 0$, and $0 < \sin s \leq 1$ on $\Theta^\circ \subset (0, \pi)$. So

$$\sin t \zeta(t) = - \sum_{s \neq t} \sin s \zeta(s) \geq - \sum_{s \neq t} \sin s e(s) \geq -E,$$

$$\sin t \zeta(t) \leq \sin t e(t) \leq E.$$

$\square$

At the pinned normals $\sin t \geq \sin \omega$; at $t = \pi/2$ the bound has no denominator.

6. THE PINNED BOUNDS AND THE RIGHT ANGLE

Baek shows that a balanced maximum sofa S_ω with $|S_\omega| \geq 11/5$ and rotation angle $\omega \in [\omega_0, \pi/2)$ has a rotated copy that turns through the full right angle [1, Thm. 1.5.2]. The proof uses the balance only through two inequalities, $w_K^\circ \leq \sigma_K(\pi/2)$ and $z_K^\circ \leq \sigma_K(\omega)$, which bound the wedge gaps on the two floors $y = 0$ and $l(\omega, 0)$ by the lengths of the sides of the cap on the opposite walls $y = 1$ and $l(\omega, 1)$ [1, Thm. 4.1.4]. We prove these

inequalities for K_* from the defect bounds of Section 5, and then follow Baek's proof with them as hypotheses. In Sections 6.1 and 6.2 we have $\omega < \pi/2$, and K_* is as in Section 4; Proposition 6.9 also covers $\omega = \pi/2$.

6.1. The pinned bounds. For a cap $K \in \mathcal{K}_\omega^c$ and $t \in (0, \omega)$ let

$$w_K(t) = h_K(0) - \frac{h_K(t) - 1}{\cos t}, \quad z_K(t) = h_K(\omega + \frac{\pi}{2}) - \frac{h_K(t + \pi/2) - 1}{\cos(\omega - t)},$$

the *wedge gaps*: with $A = (h_K(0), 0) = A_K^-(0)$ and $C = h_K(\omega + \pi/2)v_\omega = C_K^+(\omega)$ the ends of the upper boundary of K (see (2.3)), $w_K(t)$ is the distance from A to the point $W_K(t) = ((h_K(t) - 1)/\cos t, 0)$ where the inner wall $b_K(t)$ meets the x -axis, and $z_K(t)$ is the distance from C to the point where $d_K(t)$ meets $l(\omega, 0)$ [1, Def. 2.5.5]. Let $w_K^\circ = \inf_{t \in (0, \omega)} w_K(t)$ and $z_K^\circ = \inf_{t \in (0, \omega)} z_K(t)$ [1, Def. 4.1.1]. As $t \rightarrow \omega$, $w_K(t) \rightarrow h_K(0)$, so $w_K^\circ \leq h_K(0)$; as $t \rightarrow 0$, $z_K(t) \rightarrow h_K(\omega + \pi/2)$, so $z_K^\circ \leq h_K(\omega + \pi/2)$.

Lemma 6.1 (wedge gaps and the polyline). *Every polygon cap $K \in \mathcal{K}_\Theta^c$ satisfies $w_K^\circ \leq \tau_K(\pi/2)$ and $z_K^\circ \leq \tau_K(\omega)$.*

This is the first half of the proof of [1, Thm. 4.1.2].

Proof. The bottom edge $e_K(3\pi/2) = K \cap \{y = 0\}$ is the segment from O to $A = (h_K(0), 0)$, so $\sigma_K(3\pi/2) = h_K(0)$. A point $(x, 0)$ of $\mathcal{N}_\Theta(K)$ lies in F_ω , so $x \geq 0$, and in some $Q_K^-(t)$, $t \in \Theta$, so $x \cos t < h_K(t) - 1$, that is $x < h_K(0) - w_K(t) \leq h_K(0) - w_K^\circ$. Hence $\mathcal{N}_\Theta(K) \cap \{y = 0\} \subset [0, h_K(0) - w_K^\circ) \times \{0\}$, and by Fact 2.10(b)

$$\begin{aligned} h_K(0) - \tau_K(\pi/2) &= \sigma_K(3\pi/2) - \tau_K(\pi/2) \\ &= \mathcal{H}^1(\mathcal{N}_\Theta(K) \cap l(\pi/2, 0)) \leq h_K(0) - w_K^\circ, \end{aligned}$$

where we used $w_K^\circ \leq h_K(0)$. The second inequality is the same argument on the line $l(\omega, 0)$, with the bottom edge $e_K(\omega + \pi) = [O, C]$ of length $\sigma_K(\omega + \pi) = h_K(\omega + \pi/2)$, the points λv_ω of $Q_K^-(t)$, which have $\lambda < h_K(\omega + \pi/2) - z_K(t)$, and Fact 2.10(b) with $t = \omega$. $\square$

Lemma 6.2 (continuity). *For caps $K, K' \in \mathcal{K}_\omega^c$ with $d_H(K, K') = \varepsilon$, $|w_K^\circ - w_{K'}^\circ| \leq (1 + \sec \omega)\varepsilon$ and $|z_K^\circ - z_{K'}^\circ| \leq (1 + \sec \omega)\varepsilon$.*

Proof. For $t \in (0, \omega)$, $|w_K(t) - w_{K'}(t)| \leq \varepsilon + \varepsilon \sec t \leq (1 + \sec \omega)\varepsilon$; similarly for z , since $\cos(\omega - t) \geq \cos \omega$. The infima differ by at most as much. [1, Lemma 4.1.1] $\square$

Baek obtains [1, Thm. 4.1.4] from the weak convergence of surface area measures [1, Thm. 4.1.3] and the Portmanteau theorem. The following elementary lemma, which uses only support functions, replaces that step.

Lemma 6.3 (upper semicontinuity of edge lengths). *If convex bodies $K_j \rightarrow K$ in d_H , then $\limsup_j \sigma_{K_j}(t) \leq \sigma_K(t)$ for every t .*

Proof. For $0 < \delta < \pi$,

$$\begin{aligned} h_K(t + \delta) &\geq \langle v_K^+(t), u_{t+\delta} \rangle = h_K(t) \cos \delta + \langle v_K^+(t), v_t \rangle \sin \delta, \\ h_K(t - \delta) &\geq \langle v_K^-(t), u_{t-\delta} \rangle = h_K(t) \cos \delta - \langle v_K^-(t), v_t \rangle \sin \delta. \end{aligned}$$

Adding, and using $\sigma_K(t) = \langle v_K^+(t) - v_K^-(t), v_t \rangle$,

$$\sigma_K(t) \leq \Delta_K(\delta) = \frac{h_K(t + \delta) + h_K(t - \delta) - 2h_K(t) \cos \delta}{\sin \delta}.$$

$\Delta_K(\delta)$ depends continuously on the values of h_K , so $\Delta_{K_j}(\delta) \rightarrow \Delta_K(\delta)$ and $\limsup_j \sigma_{K_j}(t) \leq \Delta_K(\delta)$ for every δ . By Fact 2.1(a), $\Delta_K(\delta) \rightarrow \langle v_K^+(t) - v_K^-(t), v_t \rangle = \sigma_K(t)$ as $\delta \downarrow 0$. $\square$

Proposition 6.4 (pinned bounds). $w_{K_*}^\circ \leq \sigma_{K_*}(\pi/2)$ and $z_{K_*}^\circ \leq \sigma_{K_*}(\omega)$.

Proof. Take the selection $K_j \rightarrow K_*$ of Proposition 4.8 in $[-R_0, R_0] \times [0, 1] \subset \{\|p\| \leq R\}$, $R = R_0 + 1$, and $\eta_j = d_H(K_j, K_*) \rightarrow 0$; at every sampled normal $|h_{K_j} - h_*| \leq \eta_j$, and the total weight is $\bar{\nu} < 1$. By Corollary 5.6 at $t = \pi/2$, with $e_j = (2 + 4\gamma_\omega)\eta_j$, $|\sigma_{K_j}(\pi/2) - \tau_{K_j}(\pi/2)| \leq e_j$; so by Lemma 6.1

$$w_{K_j}^\circ - e_j \leq \tau_{K_j}(\pi/2) - e_j \leq \sigma_{K_j}(\pi/2).$$

By Lemma 6.2, $w_{K_j}^\circ \rightarrow w_{K_*}^\circ$, and by Lemma 6.3, $\limsup_j \sigma_{K_j}(\pi/2) \leq \sigma_{K_*}(\pi/2)$. This gives the first inequality. For the second use $t = \omega$, where Corollary 5.6 divides by $\sin \omega > 0$. $\square$

6.2. The triangle in the niche. Let $\omega \in [\omega_0, \pi/2)$, so that $\omega > \pi/4$. Put $c = c_\omega = \sec \omega - \tan \omega$, so that $o_\omega - v_0 = c u_0$ and $o_\omega - u_\omega = c v_\omega$ [1, Prop. 4.2.1], and let Δ_ω be the triangle with vertices O, cu_0, cv_ω , which cuts off the corner O of P_ω . Let $d_{\min} = 5/4$ if $\omega < \tan^{-1}(11/5)$ and $d_{\min} = 11/10$ otherwise, and for $d \in [0, \tan \omega]$ let $R_{\omega,d} = P_\omega \cap H_-(0, d+c) \cap H_-(\omega+\pi/2, d+c)$ be P_ω without its two acute corners.

Fact 6.5 (numerical inequalities at the corner). *Let $\omega \in [\omega_0, \pi/2)$.*

(a) $|R_{\omega,d_{\min}}| < 11/5$. [1, Lemma 4.2.2]

(b) *If $d \in [d_{\min}, \tan \omega]$, $r_y = 1 - d \cot \omega \geq 0$ and $g = \sqrt{1 - r_y^2}$, then $d \sin \omega > 1$ and $g > 2 \cos \omega$. [1, Lemma 4.2.4, in the range $[\sec^{-1}(2.2), \pi/2)$ of its proof and of its use (E9)]*

Lemma 6.6 (the triangle lies in the niche). *Let $\omega \in [\omega_0, \pi/2)$ and $K \in \mathcal{K}_\omega^c$ with $\mathcal{A}_\omega(K) \geq 11/5$, $w_K^\circ \leq \sigma_K(\pi/2)$ and $z_K^\circ \leq \sigma_K(\omega)$. Then for some $t_\sharp \in (0, \omega)$ the three vertices O, cu_0, cv_ω of Δ_ω lie in $Q_K^-(t_\sharp)$.*

This is [1, Thm. 4.2.5] for a cap that satisfies the pinned bounds, in place of a balanced maximum cap; the proof there uses balance only through them.

Proof. Step 1: a long cap. Since $|K| \geq \mathcal{A}_\omega(K) \geq 11/5 > |R_{\omega,d_{\min}}|$ and $K \subset P_\omega$, K is not contained in $R_{\omega,d_{\min}}$, so $h_K(0) \geq d_{\min} + c$ or $h_K(\omega + \pi/2) \geq d_{\min} + c$. We may assume the first. Otherwise replace K by its mirror image

K^m : it is a cap with the same sofa area, and $h_{K^m}(0) = h_K(\omega + \pi/2)$, $w_{K^m}^\circ = z_K^\circ$, $\sigma_{K^m}(\pi/2) = \sigma_K(\omega)$, so it satisfies the hypotheses; and since M_ω fixes O , exchanges cu_0 and cv_ω and maps $Q_{K^m}^-(t)$ to $Q_K^-(\omega - t)$ [1, Prop. 2.5.4], the conclusion for K^m at $t_\sharp$ is the conclusion for K at $\omega - t_\sharp$. Put $d = h_K(0) - c \geq d_{\min}$ and $q_0 = (h_K(0), 0) = A_K^-(0) \in K$. Since $K \subset P_\omega$, $h_K(0) \leq \sec \omega$, so $d \leq \tan \omega$, and $q_0 = (o_\omega - v_0) + d u_0$.

Step 2: the bottom side. The lines $l_K(0)$ and $l_K(\omega) = l(\omega, 1)$ meet at $r = (h_K(0), r_y)$ with $r_y = 1 - d \cot \omega \geq 0$ (as $q_0 \in P_\omega$). Let $g = \sqrt{1 - r_y^2}$ and $s = q_0 - g u_0$: the triangle s, q_0, r has a right angle at q_0 , and $\|r - s\| = 1$. For $t \in (0, \omega)$, u_t is a nonnegative combination of u_0 and u_ω , and $K \subset H_K(0) \cap H_K(\omega)$, so $h_K(t) \leq \langle r, u_t \rangle$, and

$$h_K(t) - 1 \leq \langle r, u_t \rangle - \langle r - s, u_t \rangle = \langle s, u_t \rangle.$$

So s lies on or to the right of $W_K(t)$, that is $w_K(t) \geq g$. Hence $g \leq w_K^\circ$.

Step 3: the top side. By hypothesis, $g \leq w_K^\circ \leq \sigma_K(\pi/2)$. The point o_ω lies in K , and it is the right end of the top edge $e_K(\pi/2) = K \cap \{y = 1\}$, because $K \subset P_\omega$. So $q_1 = o_\omega - g u_0$ lies on $e_K(\pi/2)$, and $q_1 \in K$.

Step 4: the hallway of angle $t_\sharp = \pi/2 - \omega$. Note $t_\sharp \in (0, \pi/4) \subset (0, \omega)$. Let $X = \{q_0, q_1\} \subset K$; then $h_X \leq h_K$, so $Q_X^-(t_\sharp) \subset Q_K^-(t_\sharp)$. We have $q_1 - q_0 = v_0 - (d + g)u_0 = -\alpha u_{t_\sharp} + \beta v_{t_\sharp}$ with $\alpha = (d + g) \cos t_\sharp - \sin t_\sharp \geq 0$ and $\beta = \cos t_\sharp + (d + g) \sin t_\sharp \geq 0$, since $d + g \geq 1 > \tan t_\sharp$. So q_0 maximizes $\langle \cdot, u_{t_\sharp} \rangle$ and q_1 maximizes $\langle \cdot, v_{t_\sharp} \rangle$ on X , and

$$\begin{aligned} Q_X^-(t_\sharp) &= \{p : \langle p, u_{t_\sharp} \rangle < m_1, \langle p, v_{t_\sharp} \rangle < m_2\}, \\ m_1 &= \langle q_0, u_{t_\sharp} \rangle - 1, \quad m_2 = \langle q_1, v_{t_\sharp} \rangle - 1. \end{aligned}$$

Step 5: the vertices. By Fact 6.5(b), $d \sin \omega > 1$ and $g > 2 \cos \omega$. Now $\langle q_0 - cu_0, u_{t_\sharp} \rangle = d \sin \omega > 1$, so $m_1 > c \cos t_\sharp > 0$; and $\langle q_1 - cv_\omega, v_{t_\sharp} \rangle = \langle u_\omega - gu_0, v_{t_\sharp} \rangle = -\cos 2\omega + g \cos \omega > 1$ (this is $g > 2 \cos \omega$, as $1 + \cos 2\omega = 2 \cos^2 \omega$), so $m_2 > c \cos(\omega - t_\sharp) > 0$. Then O satisfies both inequalities. For cu_0 : $\langle cu_0, u_{t_\sharp} \rangle = c \cos t_\sharp < m_1$ and $\langle cu_0, v_{t_\sharp} \rangle = -c \sin t_\sharp < 0 < m_2$. For cv_ω : $\langle cv_\omega, u_{t_\sharp} \rangle = -c \sin(\omega - t_\sharp) < 0 < m_1$ and $\langle cv_\omega, v_{t_\sharp} \rangle = c \cos(\omega - t_\sharp) < m_2$. So the three points lie in $Q_X^-(t_\sharp) \subset Q_K^-(t_\sharp)$. $\square$

6.3. A rotated copy turns through the right angle.

Lemma 6.7 (the width of the sofa). *Let $\omega \in (0, \pi/2)$ and $K \in \mathcal{K}_\omega^c$, and suppose that $O, cu_0, cv_\omega \in Q_K^-(t_\sharp)$ for some $t_\sharp \in (0, \omega)$. Then*

- (a) *every point $\lambda u_0 + \mu v_\omega$ with $\lambda, \mu \geq 0$ and $\lambda + \mu < c$ lies in $\mathcal{N}(K)$;*
- (b) *$\langle p - q, u_t \rangle \leq 1$ for all $p \in K$, $q \in K \setminus \mathcal{N}(K)$ and $t \in [\omega, \pi/2]$.*

Proof. (a) Such a point is a convex combination of O , cu_0 and cv_ω with the weights $1 - (\lambda + \mu)/c > 0$, λ/c , μ/c . The open convex set $Q_K^-(t_\sharp)$ contains all three vertices, hence the point, and the point lies in F_ω since $\langle u_0, u_\omega \rangle, \langle v_\omega, u_\omega \rangle, \langle u_0, u_{\pi/2} \rangle, \langle v_\omega, u_{\pi/2} \rangle \geq 0$.

(b) Write $q = \lambda u_0 + \mu v_\omega$. As $q \in P_\omega$, $\lambda = \langle q, u_\omega \rangle / \cos \omega \geq 0$ and $\mu = q_y / \cos \omega \geq 0$, and $\lambda + \mu \geq c$ by (a). For $t \in [\omega, \pi/2]$ both $\cos t$ and $\sin(t - \omega)$ are nonnegative, so $\langle q, u_t \rangle = \lambda \cos t + \mu \sin(t - \omega) \geq c \min(\cos t, \sin(t - \omega))$, while $\langle p, u_t \rangle \leq \langle o_\omega, u_t \rangle = c \cos t + \sin t$ as o_ω is the vertex of P_ω farthest in these directions. If $\cos t \leq \sin(t - \omega)$, then $\langle p - q, u_t \rangle \leq \sin t \leq 1$. Otherwise, using $c(1 + \sin \omega) = \cos \omega$ and $c \cos \omega = 1 - \sin \omega$,

$$\langle p - q, u_t \rangle \leq c \cos t + \sin t - c \sin(t - \omega) = \cos(t - \omega) \leq 1. \quad \square$$

Lemma 6.8 (turning inside the horizontal side). *Let S be a moving sofa with rotation angle $\omega \in (0, \pi/2)$ such that $\langle p - q, u_t \rangle \leq 1$ for all $p, q \in S$ and $t \in [\omega, \pi/2]$. Then $R_\psi S$, $\psi = \pi/2 - \omega$, is a moving sofa with rotation angle $\pi/2$.*

Proof. Let $(\theta, \mathbf{c})$ be a movement of S with rotation angle ω . For $\rho \in [0, \psi]$ the height of $R_\rho S$, that is its width in the direction $u_{\pi/2}$, is the width of S in the direction $u_{\pi/2-\rho}$ with $\pi/2 - \rho \in [\omega, \pi/2]$, so it is at most 1. Hence the translations $b(\rho) = (1 - \max_{p \in S}(R_\rho p)_x, -\min_{p \in S}(R_\rho p)_y)$, which depend continuously on ρ , put $R_\rho S + b(\rho)$ in H_L . The movement of $R_\psi S$ has three phases: turn clockwise by ψ through the positions $R_\rho S + b(\rho)$, ρ from ψ down to 0; translate along a segment in H_L , which is convex, from $S + b(0)$ to the starting position $S + \mathbf{c}(0)$ of the movement of S ; and follow that movement. The total rotation is $\psi + \omega = \pi/2$, and $R_\psi S$ is closed and connected. $\square$

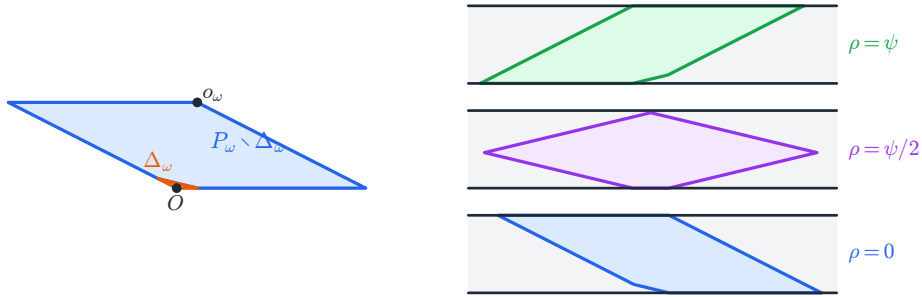


FIGURE 5. Lemmas 6.7 and 6.8, for $\omega = 1.1$. Left: the parallelogram P_ω (dashed), its corner triangle Δ_ω (orange) and the pentagon $P_\omega \setminus \Delta_\omega$ (blue); the sofa $K \setminus \mathcal{N}(K)$ lies in the pentagon, up to the far side of Δ_ω . Right: the pentagon turned by $\rho = \psi$, $\psi/2$ and 0, $\psi = \pi/2 - \omega$, in a horizontal strip of width one: it fits for every $\rho \in [0, \psi]$, since its width in the direction u_t is at most 1 for $t \in [\omega, \pi/2]$.

Proposition 6.9 (a right-angle monotone sofa). *Let $\omega \in [\omega_0, \pi/2]$ and let T be a monotone sofa with rotation angle ω , $|T| = |G|$, and cap $K = \mathcal{C}(T)$, as in Proposition 3.2. Then $R_\psi T$, $\psi = \pi/2 - \omega$, is a moving sofa with rotation*

angle $\pi/2$. Consequently there are $w_1 \in \mathbb{R}^2$ and a monotone sofa U with rotation angle $\pi/2$ such that

$$R_\psi T + w_1 \subset U, \quad |U| = |G|,$$

and the cap $\mathcal{C}(U) \in \mathcal{K}_{\pi/2}^c$ is maximizing, with $\mathcal{A}(\mathcal{C}(U)) = |G|$.

Proof. If $\omega = \pi/2$, $\psi = 0$ and T is a moving sofa with rotation angle $\pi/2$. Let $\omega < \pi/2$. The cap K is maximizing with $\mathcal{A}_\omega(K) = |G| \geq 11/5$, so Proposition 6.4 applies to $K_* = K$ and gives the pinned bounds, Lemma 6.6 puts the vertices of Δ_ω in some $Q_K^-(t_\#)$, and Lemma 6.7(b) gives $\langle p-q, u_t \rangle \leq 1$ for $p, q \in T = K \setminus \mathcal{N}(K)$ and $t \in [\omega, \pi/2]$. By Lemma 6.8, $R_\psi T$ is a moving sofa with rotation angle $\pi/2$, and $|R_\psi T| = |G|$. The second part is Proposition 3.2 for the moving sofa $S' = R_\psi T$. $\square$

By Propositions 3.2 and 6.9, there are $w_0, w_1 \in \mathbb{R}^2$ such that, with $g_1(p) = R_\psi(p + w_0) + w_1$,

$$g_1(S) \subset U, \quad |U| = |G|, \quad (6.1)$$

where U is a monotone sofa with rotation angle $\pi/2$ whose cap is maximizing.

7. CURVATURE BOUNDS AND THE INJECTIVITY CONDITION

Baek proves the injectivity condition for a balanced maximum cap K with rotation angle $\pi/2$ from the inequality $\sigma_K \leq k_0(g_K^+(t))dt$ on $[0, \pi/2]$ [1, Thm. 6.4.3], which is the limit of $\sigma_{K_n}(t) \leq k_0(g_{K_n}^+(t))\delta + O(\delta^2)$ for maximum polygon caps K_n with step size δ [1, Thm. 6.3.3]; the proof of the latter uses the balance $\sigma = \tau$. For the polygon caps of Section 4 the defect bound of Proposition 5.2 replaces the balance, and its errors add up to at most 2η by (4.1). In this section we use Sections 4 and 5 with $\omega = \pi/2$ and $\mathcal{A} = \mathcal{A}_{\pi/2}$: $K_* \in \mathcal{K}_{\pi/2}^c$ is a maximizing cap with $\mathcal{A}(K_*) > 0$, $\Theta_k = \{j\delta : 0 < j < n\}$ with $n = 2^{k+1}$ and step size $\delta = \delta_k = \pi/(2n) \leq \pi/4$. All the normals of Θ_k are floating. A polygon cap $K \in \mathcal{K}_k^c$ has no edge with normal angle 0 or π and none with normal in $(-\pi/2, 0)$; its edge normals are among $\delta, 2\delta, \dots, (2n-1)\delta$ and $3\pi/2$. For an angle u let $H_K^b(u)$ and $H_K^d(u)$ be as in Section 2.5, so that $Q_K^-(u)$ is the complement of $H_K^b(u) \cup H_K^d(u)$.

7.1. One normal of a polygon cap.

Lemma 7.1 (the inner wall near the corner). *Let $K \in \mathcal{K}_k^c$, $t \in \Theta_k$, $q = \tan(\delta/2)$, and $\mathbf{x} = \mathbf{x}_K(t)$. Let $\vec{b} = \vec{b}_K(t) = \{\mathbf{x} - sv_t : s \geq 0\}$. Then the set of $s \geq 0$ with $\mathbf{x} - sv_t \in H_K^d(t - \delta)$ is $[0, \alpha_-]$ and the set of $s \geq 0$ with $\mathbf{x} - sv_t \in H_K^d(t + \delta)$ is $[0, \alpha_+]$, where*

$$\alpha_- = \tan \delta (g_K^-(t) - 1 + q), \quad \alpha_+ = \tan \delta (1 - g_K^+(t) + q)$$

(an interval is empty if its right end is negative); and the set of $s \in \mathbb{R}$ with $\mathbf{x} - sv_t \in H_K^b(t - \delta) \cap H_K^b(t + \delta)$ is an interval, empty or of length $2q - \sigma_K(t)$.

This is [1, Lemma 6.3.2], together with the length $2 \tan(\delta/2) - \sigma_K(t)$ that Baek asserts in the proof of [1, Thm. 6.3.3]. They are stated there for maximum polygon caps, but the proofs use only that no normal angle of K lies in $(t - \delta, t) \cup (t, t + \delta) \cup (t + \frac{\pi}{2} - \delta, t + \frac{\pi}{2}) \cup (t + \frac{\pi}{2}, t + \frac{\pi}{2} + \delta)$, which holds for every $K \in \mathcal{K}_k^c$.

Proof. Since no normal angle of K lies between $t + \pi/2$ and $t + \delta + \pi/2$, the supporting lines $l_K(t + \pi/2)$ and $l_K(t + \delta + \pi/2)$ meet at a vertex r of K , namely $r = C_K^+(t) = \mathbf{y}_K(t) - g_K^+(t)u_t$, and $h_K(t + \delta + \pi/2) = \langle r, v_{t+\delta} \rangle$. As $\mathbf{x} - r = (g_K^+(t) - 1)u_t - v_t$,

$$\begin{aligned} \langle \mathbf{x}, v_{t+\delta} \rangle - h_K(t + \delta + \frac{\pi}{2}) + 1 &= \langle \mathbf{x} - r, v_{t+\delta} \rangle + 1 \\ &= -(g_K^+(t) - 1) \sin \delta - \cos \delta + 1. \end{aligned}$$

Now $\langle \mathbf{x} - sv_t, v_{t+\delta} \rangle = \langle \mathbf{x}, v_{t+\delta} \rangle - s \cos \delta$, so $\mathbf{x} - sv_t \in H_K^d(t + \delta)$ iff $s \leq \alpha_+ = [(1 - g_K^+(t)) \sin \delta + 1 - \cos \delta] / \cos \delta$, and $(1 - \cos \delta) / \cos \delta = \tan \delta \tan(\delta/2)$. For $t - \delta$ use $r' = C_K^-(t) = \mathbf{y}_K(t) - g_K^-(t)u_t$ in the same way. Next, $\langle \mathbf{x} - sv_t, u_{t+\delta} \rangle = \langle \mathbf{x}, u_{t+\delta} \rangle \pm s \sin \delta$, so the conditions $\langle p, u_{t-\delta} \rangle \geq h_K(t - \delta) - 1$ and $\langle p, u_{t+\delta} \rangle \geq h_K(t + \delta) - 1$ on $p = \mathbf{x} - sv_t$ read $s \geq s_-$ and $s \leq s_+$ with

$$\begin{aligned} s_+ - s_- &= \frac{2 \cos \delta (h_K(t) - 1) - h_K(t + \delta) - h_K(t - \delta) + 2}{\sin \delta} \\ &= \frac{2(1 - \cos \delta) - \sigma_K(t) \sin \delta}{\sin \delta} = 2q - \sigma_K(t), \end{aligned}$$

where we used $u_{t+\delta} + u_{t-\delta} = 2 \cos \delta u_t$, $\langle \mathbf{x}, u_t \rangle = h_K(t) - 1$ and, since no normal angle of K lies in $(t - \delta, t) \cup (t, t + \delta)$, $h_K(t + \delta) + h_K(t - \delta) - 2 \cos \delta h_K(t) = \sigma_K(t) \sin \delta$ (equality in the inequality in the proof of Lemma 6.3). $\square$

Lemma 7.2 (the polygon bound at one normal). *For $K \in \mathcal{K}_k^c$ and $t \in \Theta_k$,*

$$\begin{aligned} \tau_K(t) &\leq \max(\tan \delta (g_K^-(t) - 1 + q), \tan \delta (1 - g_K^+(t) + q), 0) \\ &\quad + \max(2q - \sigma_K(t), 0), \quad q = \tan \frac{\delta}{2}. \end{aligned}$$

Proof. Let $X = \partial \mathcal{N}_{\Theta_k}(K) \cap \vec{b}_K(t)$; by Fact 2.10(a), $\tau_K(t) = \mathcal{H}^1(X)$. As $X \subset \text{cl} \mathcal{N}_{\Theta_k}(K) \subset \{y \geq 0\}$ and the line $b_K(t)$ is not horizontal, at most one point of X lies on the x -axis. Let $p \in X$ with $p_y > 0$. The set $\mathcal{N}_{\Theta_k}(K)$ is open in $\{y \geq 0\}$, being a union of open quarter-planes intersected with that half-plane; so the boundary point p is not in it, and $p \notin Q_K^-(u)$ for every $u \in \Theta_k$. Also $Q_K^-(0)$ and $Q_K^-(\pi/2)$ lie in $\{y < 0\}$ (the inner corners $\mathbf{x}_K(0)$ and $\mathbf{x}_K(\pi/2)$ are on the x -axis). As $t - \delta, t, t + \delta \in \Theta_k \cup \{0, \pi/2\}$, $p \notin Q_K^-(u)$ for $u \in I = \{t - \delta, t, t + \delta\}$; so either $p \in H_K^d(u)$ for some $u \in I$, or $p \in H_K^b(u)$ for all $u \in I$. On $\vec{b}_K(t)$, $H_K^d(t)$ contains only the corner $\mathbf{x}$, and $H_K^b(t)$ everything. By Lemma 7.1, the points of the first kind form a set of length at most $\max(0, \alpha_-, \alpha_+)$, those of the second kind a set of length at most $\max(0, 2q - \sigma_K(t))$. $\square$

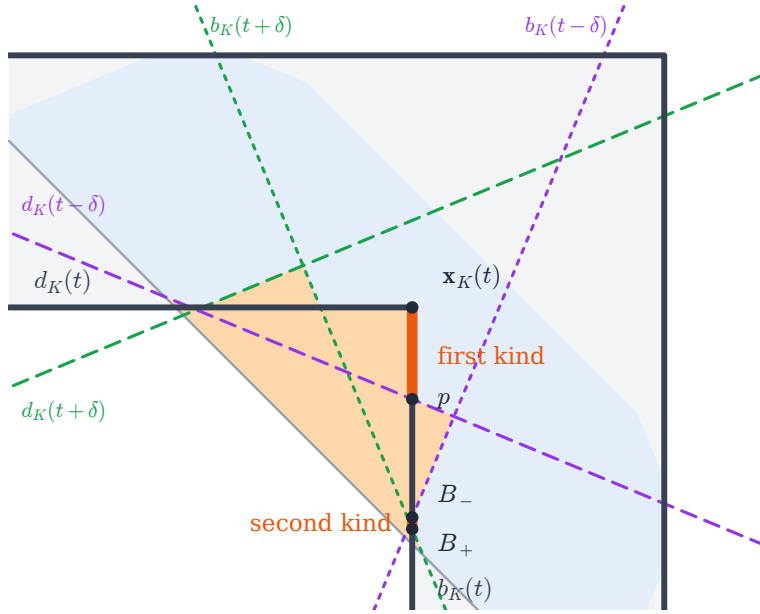


FIGURE 6. Lemmas 7.1 and 7.2, for the polygon cap with the angle set Θ_1 ($n = 4$, $\delta = \pi/8$) and the support values of Gerver's cap, at $t = \pi/4$, drawn in the frame of the supporting hallway $L_K(t)$ (the polygon niche is shaded). The part of $\vec{b}_K(t)$ above the dashed line $d_K(t - \delta)$ is the segment $[p, \mathbf{x}_K(t)]$ (points of the first kind); further down $\vec{b}_K(t)$ runs between the dotted lines $b_K(t - \delta)$ and $b_K(t + \delta)$, from B_- to B_+ (points of the second kind). The side of the niche on $\vec{b}_K(t)$ lies in the union of the two segments.

Lemma 7.3 (with a defect). *Let $K \in \mathcal{K}_k^c$, $t \in \Theta_k$, $e \geq 0$, $D \geq 0$ with $g_K^+(t) \leq D$ and $\sigma_K(t) \leq \tau_K(t) + e$. Then*

$$\sigma_K(t) \leq k_0(g_K^+(t))\delta + (D + 4)\delta^2 + e.$$

Proof. Let $M = |g_K^+(t) - 1| \leq D + 1$. As $0 \leq g_K^- \leq g_K^+$, each of $g_K^- - 1 + q$ and $1 - g_K^+ + q$ is at most $M + q$, so the first maximum in Lemma 7.2 is at most $\tan \delta (M + q) \leq \delta M + (D + 3)\delta^2$, using $\tan \delta \leq \delta + \delta^2$, $q \leq \delta$ and $\delta \leq 1$. If $\sigma_K(t) \geq 2q$, the second maximum is 0 and $\sigma_K(t) \leq \tau_K(t) + e \leq \delta M + (D + 3)\delta^2 + e$. If $\sigma_K(t) < 2q$, then $\sigma_K(t) \leq \delta M + (D + 3)\delta^2 + (2q - \sigma_K(t)) + e$ and $2q \leq \delta + \delta^2$, so $\sigma_K(t) \leq \frac{M+1}{2}\delta + \frac{D+4}{2}\delta^2 + \frac{e}{2}$. Both M and $(M + 1)/2$ are at most $k_0(g_K^+(t))$. $\square$

7.2. From normals to intervals.

Lemma 7.4 (arms between two steps). *Let $K \in \mathcal{K}_k^c$ have diameter at most D , and let $t \in \{0\} \cup \Theta_k$. Then g_K^+ is nonincreasing on $[t, t + \delta)$, and $g_K^+(t) - g_K^+(u) \leq D\delta$ for $u \in (t, t + \delta)$.*

This is [1, Lemma 6.4.1], with D in place of the bound 5 for maximum polygon caps. The half-open interval is needed: g_K^+ jumps up at $t + \delta$, by $\sigma_K(t + \delta + \pi/2)$, and Baek's statement compares $g_K^+(t)$ with $g_K^-(t + \delta)$.

Proof. Since there is no normal angle of K strictly between t and $t + \delta$, nor between $t + \pi/2$ and $t + \delta + \pi/2$, there are vertices A and C of K with $A_K^\pm(s) = A$ and $C_K^\pm(s) = C$ for $s \in (t, t + \delta)$, and also $A_K^+(t) = A$, $C_K^+(t) = C$. As $\mathbf{y}_K(s)$ lies on $l_K(s) \ni A$, we get $g_K^+(s) = \langle A - C, u_s \rangle$ on $[t, t + \delta)$. Its derivative is $\langle A - C, v_s \rangle$. It is at most 0, because $\langle A, v_s \rangle \leq h_K(s + \pi/2) = \langle C, v_s \rangle$, and at least $-\|A - C\| \geq -D$. $\square$

Lemma 7.5 (summing the steps). *Let $K \in \mathcal{K}_k^c$ have diameter at most D with $k_0(g_K^+) \leq B$ on $[0, \pi/2]$, and suppose that for some $C \geq 0$ and every $t \in \{0\} \cup \Theta_k$*

$$\sigma_K(t) \leq k_0(g_K^+(t))\delta + C\delta^2 + e(t), \quad e(t) \geq 0.$$

Then for $-\pi/2 \leq a < b \leq \pi/2$ with $a' = \max(a, 0) \leq b$,

$$\sigma_K((a, b)) \leq \int_{a'}^b k_0(g_K^+(u))du + \left(B + \frac{\pi}{2}(C + D)\right)\delta + \sum_{j=0}^{n-1} e(j\delta).$$

Proof. On $(-\pi/2, \pi/2)$, σ_K is carried by the points $j\delta$, $1 \leq j \leq n - 1$. For $t = j\delta$, $0 \leq j \leq n - 1$, Lemma 7.4, together with k_0 being 1-Lipschitz, gives $k_0(g_K^+(t))\delta \leq \int_t^{t+\delta} k_0(g_K^+) + D\delta^2$, so $\sigma_K(t) \leq \int_t^{t+\delta} k_0(g_K^+) + (C + D)\delta^2 + e(t)$. Sum over the grid points $t \in [a', b]$; they are at most n in number, their cells $[t, t + \delta)$ lie in $[a', b + \delta)$ and the part beyond b is shorter than δ , where $k_0(g_K^+) \leq B$. Also $n(C + D)\delta^2 = \frac{\pi}{2}(C + D)\delta$. $\square$

The next two lemmas are the passage to the limit in the proof of [1, Thm. 6.4.3]. Baek gets the lower semicontinuity of the mass of an open interval from the weak convergence of σ_{K_n} and the Portmanteau theorem; Lemma 7.6 replaces this by a direct argument with support functions.

Lemma 7.6 (lower semicontinuity of the mass of an open interval). *If convex bodies $K_j \rightarrow K$ in d_H and $a < b$, then $\sigma_K((a, b)) \leq \liminf_j \sigma_{K_j}((a, b))$.*

Proof. For $0 < \varepsilon < \pi$ let $\Phi_K(\varepsilon) = \langle v_K(b - \varepsilon, b), v_b \rangle - \langle v_K(a, a + \varepsilon), v_a \rangle + \int_a^b h_K$. A point $p = h_K(b)u_b + mv_b$ of $e_K(b)$ lies in $H_K(b - \varepsilon)$, i.e. $h_K(b) \cos \varepsilon - m \sin \varepsilon \leq h_K(b - \varepsilon)$, so $\langle v_K(b - \varepsilon, b), v_b \rangle \geq \langle v_K(b - \varepsilon, b), v_b \rangle$; likewise $\langle v_K^+(a), v_a \rangle \leq \langle v_K(a, a + \varepsilon), v_a \rangle$. By (2.1), $\Phi_K(\varepsilon) \leq \sigma_K((a, b))$, and $\Phi_K(\varepsilon) \rightarrow \sigma_K((a, b))$ as $\varepsilon \downarrow 0$ by Fact 2.1(a). For fixed ε , $\Phi_{K_j}(\varepsilon) \rightarrow \Phi_K(\varepsilon)$, as Φ is an explicit continuous function of the support function. Hence $\sigma_K((a, b)) = \lim_\varepsilon \lim_j \Phi_{K_j}(\varepsilon) \leq \liminf_j \sigma_{K_j}((a, b))$. $\square$

Lemma 7.7 (passing to the limit). *Let K_j be polygon caps with rotation angle $\pi/2$ that converge in d_H to $K \in \mathcal{K}_{\pi/2}^c$, and let $\varepsilon_j \rightarrow 0$ be such that, for all $-\pi/2 \leq a < b \leq \pi/2$ with $a' = \max(a, 0) \leq b$,*

$$\sigma_{K_j}((a, b)) \leq \int_{a'}^b k_0(g_{K_j}^+) + \varepsilon_j.$$

Then $\sigma_K(E) \leq \int_E k_0(g_K^+(t))dt$ for every Borel set $E \subset [0, \pi/2)$.

Proof. By Fact 2.14(a) and since k_0 is 1-Lipschitz, $\int_{a'}^b k_0(g_{K_j}^+) \rightarrow \int_{a'}^b k_0(g_K^+)$. With Lemma 7.6, $\sigma_K((a, b)) \leq \int_{a'}^b k_0(g_K^+)$ for all such $a < b$. A nonempty relatively open interval of $[0, \pi/2)$ is (a, b) with $a \geq 0$, or $[0, b) \subset (a, b)$ for every $a \in [-\pi/2, 0)$, so the inequality $\sigma_K(E) \leq \int_E k_0(g_K^+)$ holds for every such interval E , hence for every relatively open set, and by outer regularity of the finite measure $k_0(g_K^+)dt$ for every Borel set. $\square$

7.3. The curvature bounds.

Definition 7.8 (curvature bounds). A cap $K \in \mathcal{K}_{\pi/2}^c$ satisfies the *curvature bounds* if, as measures,

$$\sigma_K \leq k_0(g_K^+(t))dt \text{ on } [0, \pi/2), \quad \sigma_K \leq k_0(f_K^-(t - \pi/2))dt \text{ on } (\pi/2, \pi].$$

Proposition 7.9 (maximizing caps satisfy the curvature bounds). *Every maximizing cap $K_* \in \mathcal{K}_{\pi/2}^c$ with $\mathcal{A}(K_*) > 0$ satisfies the curvature bounds.*

Proof. Take the selection $K_j \rightarrow K_*$ of Proposition 4.8, with stages k_j , step sizes $\delta_j \rightarrow 0$, in $[-R_0, R_0] \times [0, 1]$, and $\eta_j = d_H(K_j, K_*) \rightarrow 0$. The diameter of K_j is at most $D = 2R_0 + 1$; so $g_{K_j}^+ \leq D$ and $k_0(g_{K_j}^+) \leq B = D + 1$. For $t \in \Theta_{k_j}$, Proposition 5.2 gives $\sigma_{K_j}(t) \leq \tau_{K_j}(t) + e_j(t)$ with $e_j(t) = 2\eta_j \nu_{k_j}(t) \geq 0$, so by Lemma 7.3

$$\sigma_{K_j}(t) \leq k_0(g_{K_j}^+(t))\delta_j + (D + 4)\delta_j^2 + e_j(t).$$

At $t = 0$ this holds trivially, since $\sigma_{K_j}(0) = 0$ (put $e_j(0) = 0$). By Lemma 7.5 with $C = D + 4$, for $-\pi/2 \leq a < b \leq \pi/2$ with $a' \leq b$,

$$\sigma_{K_j}((a, b)) \leq \int_{a'}^b k_0(g_{K_j}^+) + (B + \pi(D + 2))\delta_j + 2\eta_j,$$

since the weights at distinct normals add up to less than 1, (4.1), so $\sum_t e_j(t) \leq 2\eta_j$. By Lemma 7.7, K_* satisfies the first curvature bound. The reflection K_*^m is a cap with the same sofa area (Fact 2.4(d)), hence maximizing with positive sofa area; by Fact 2.13(c) and $\sigma_{K^m}(E) = \sigma_K(\pi - E)$, its first bound is the second bound for K_* . $\square$

7.4. The injectivity condition.

Lemma 7.10 (integral inequalities). *Let $K \in \mathcal{K}_{\pi/2}^c$ satisfy the curvature bounds. Then K satisfies condition (1) of Definition 2.12, and*

$$\begin{aligned} f_K(t) - 1 &\geq \int_0^t m_0(g_K(u))du \quad (0 \leq t < \frac{\pi}{2}), \\ g_K(t) - 1 &\geq \int_t^{\pi/2} m_0(f_K(u))du \quad (0 < t \leq \frac{\pi}{2}). \end{aligned}$$

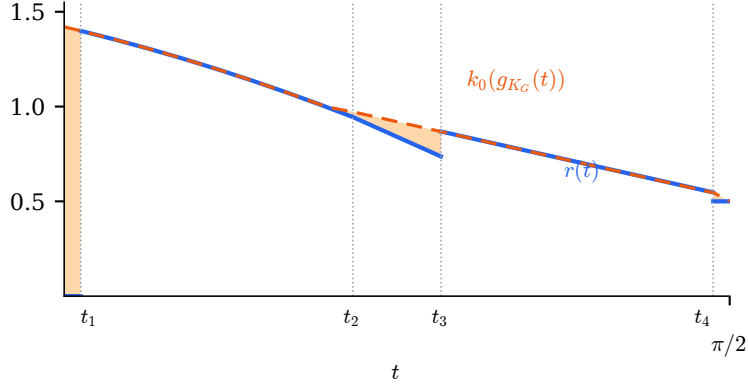


FIGURE 7. The first curvature bound for Gerver's cap, which maximizes $\mathcal{A}$ (Fact 2.7): the density $r(t)$ of σ_{K_G} on $[0, \pi/2]$ (blue) lies below $k_0(g_{K_G}(t))$ (orange, dashed), computed from Romik's rotation path. They agree on $[t_1, t_*]$, where $g_{K_G}(t_*) = 2$ ($t_* = 0.616\dots$), and on $[t_3, t_4]$; the gap is shaded.

Proof. Each restriction of σ_K is dominated by a finite measure with a density, so it has a density (Radon–Nikodym) and condition (1) holds. By Fact 2.13(d), f_K, g_K are defined and continuous, $f_K(0) = 1$, and by Fact 2.13(b), for $t < \pi/2$, $f_K(t) - 1 = \int_0^t g_K - \sigma_K((0, t]) \geq \int_0^t (g_K - k_0(g_K)) = \int_0^t m_0(g_K)$. For the second inequality apply the first to the reflection K^m , which satisfies the curvature bounds with their roles exchanged, and use Fact 2.13(c): $f_{K^m}(s) = g_K(\pi/2 - s)$ and $g_{K^m}(s) = f_K(\pi/2 - s)$, with $t = \pi/2 - s$. $\square$

Lemma 7.11 (the lower sequence). *Let $f, g : [0, \pi/2] \rightarrow [0, \infty)$ be continuous with $f(t) - 1 \geq \int_0^t m_0(g)$ for $t \in [0, \pi/2)$ and $g(t) - 1 \geq \int_t^{\pi/2} m_0(f)$ for $t \in (0, \pi/2]$. Then for every $j \geq 0$, $f_j(t) \leq f(t)$ on $[0, \pi/2)$ and $f_j(\pi/2 - t) \leq g(t)$ on $(0, \pi/2]$.*

Proof. Induction on j , the case $j = 0$ being $f_0 = 0 \leq f, g$. If it holds for j , then for $t \in [0, \pi/2)$, since m_0 is nondecreasing and $f_j(\pi/2 - u) \leq g(u)$ for $u \in (0, t]$, $\mathcal{F}f_j(t) = 1 + \int_0^t m_0(f_j(\pi/2 - u))du \leq 1 + \int_0^t m_0(g) \leq f(t)$, and with $f_j \leq f$ we get $f_{j+1} = \max(f_j, \mathcal{F}f_j) \leq f$ on $[0, \pi/2)$. For $t \in (0, \pi/2]$, $\mathcal{F}f_j(\pi/2 - t) = 1 + \int_t^{\pi/2} m_0(f_j(s))ds \leq 1 + \int_t^{\pi/2} m_0(f) \leq g(t)$, and with $f_j(\pi/2 - t) \leq g(t)$ this gives $f_{j+1}(\pi/2 - t) \leq g(t)$. $\square$

Proposition 7.12 (injectivity). *Every $K \in \mathcal{K}_{\pi/2}^c$ that satisfies the curvature bounds satisfies the injectivity condition, and $f_K, g_K > 1$ on $(0, \pi/2)$.*

Proof. By Lemma 7.10, K satisfies (1) and the integral inequalities, and f_K, g_K are continuous and nonnegative. By Lemma 7.11 and Fact 2.14(b), for $t \in (0, \pi/2)$, $1 < f_{11}(t) \leq f_K(t)$ and $1 < f_{11}(\pi/2 - t) \leq g_K(t)$. By Fact 2.13(d), $\mathbf{x}_K$ is continuously differentiable with $\mathbf{x}'_K = -(f_K - 1)u_t + (g_K - 1)v_t$, which gives (2) and (3). $\square$

Corollary 7.13 (maximizing right-angle caps are in $\mathcal{K}^i$). *Every $K \in \mathcal{K}_{\pi/2}^c$ with $\mathcal{A}(K) = |G|$ lies in $\mathcal{K}^i$.*

Proof. K is maximizing (Fact 2.7) and $\mathcal{A}(K) = |G| > 0$, so it satisfies the curvature bounds (Proposition 7.9) and the injectivity condition (Proposition 7.12); and $|K| \geq \mathcal{A}(K) = |G| \geq 11/5$. $\square$

8. EQUALITY IN BAEK'S UPPER BOUND

Let $K \in \mathcal{K}_{\pi/2}^c$ with $\mathcal{A}(K) = |G|$. By Corollary 7.13, $K \in \mathcal{K}^i$. Write $K_0 = K_G$ for Gerver's cap, $K_1 = K$, $h_i = h_{K_i}$ and $\Delta = h_1 - h_0$; since both are caps with rotation angle $\pi/2$, $\Delta(\pi/2) = 0$. Let $\xi_0 = \xi_G$ and $\xi_1 = \xi_K$ be the canonical triples (Section 2.5), and $K_{1/2}$, $B_{1/2}$, $D_{1/2}$ the components of $\xi_{1/2} = \frac{1}{2}\xi_0 + \frac{1}{2}\xi_1$.

8.1. Equality in the concavity of $\mathcal{Q}$.

Lemma 8.1 (equality in $\mathcal{S}$). *For each of the four terms $\mathcal{M}_K(a, b; \mathbf{z}_K)$ of $\mathcal{S}_K$,*

$$\mathcal{M}_{K_{1/2}}(a, b; \mathbf{z}_{K_{1/2}}) = \frac{1}{2}\mathcal{M}_{K_0}(a, b; \mathbf{z}_{K_0}) + \frac{1}{2}\mathcal{M}_{K_1}(a, b; \mathbf{z}_{K_1}).$$

Proof. By Fact 2.15(a), (c), $|G| = \mathcal{A}(K) \leq \mathcal{Q}(\xi_1) \leq |G|$, and $\mathcal{Q}(\xi_0) = |G|$, so $\mathcal{Q}(\xi_0) = \mathcal{Q}(\xi_1) = |G|$. The set $\mathcal{T}$ is closed under Minkowski combinations, so $\xi_{1/2} \in \mathcal{T}$. By Fact 2.15(b), $\mathcal{Q} = \Lambda - \mathcal{S} - \mathcal{R} - \mathcal{L}$ with Λ convex-linear and the three terms convex, so $\mathcal{Q}(\xi_{1/2}) \geq \frac{1}{2}\mathcal{Q}(\xi_0) + \frac{1}{2}\mathcal{Q}(\xi_1) = |G|$, while $\mathcal{Q}(\xi_{1/2}) \leq |G|$ by Fact 2.15(c). So equality holds, and since Λ is convex-linear this says

$$[\mathcal{S}_{K_{1/2}} - \frac{1}{2}\mathcal{S}_{K_0} - \frac{1}{2}\mathcal{S}_{K_1}] + [\mathcal{R}_{B_{1/2}} - \frac{1}{2}\mathcal{R}_{B_0} - \frac{1}{2}\mathcal{R}_{B_1}] + [\mathcal{L}_{D_{1/2}} - \frac{1}{2}\mathcal{L}_{D_0} - \frac{1}{2}\mathcal{L}_{D_1}] = 0.$$

Each bracket is ≤ 0 by convexity, so each is 0. The first is the sum of the corresponding brackets of the four terms of $\mathcal{S}$, which are all ≤ 0 ; so each of them is 0. $\square$

8.2. Equal tangent displacements. By Fact 2.15(b), for each term the displacement $\varrho_{K, \mathbf{z}_K}(t)$ is convex-linear in K . Write $\varrho_i = \varrho_{K_i, \mathbf{z}_{K_i}}$, so that $\varrho_{1/2} = \frac{1}{2}(\varrho_0 + \varrho_1)$ and

$$\frac{1}{2}\varrho_0^2 + \frac{1}{2}\varrho_1^2 - \varrho_{1/2}^2 = \frac{1}{4}(\varrho_0 - \varrho_1)^2.$$

So the equality of Lemma 8.1 reads

$$\frac{1}{8} \int_a^b (\varrho_0(t) - \varrho_1(t))^2 dt = 0, \tag{8.1}$$

and $\varrho_0 = \varrho_1$ almost everywhere on $[a, b]$. The four intervals are $(0, \varphi)$, $(\varphi, \pi/2 - \varphi)$, $(\pi/2 - \varphi, \pi/2)$ and $(\pi/2, \pi)$; none contains $\pi/2$. On each of them both K_i satisfy condition (1) of the injectivity condition, so by Fact 2.13(d) the vertices $v_{K_i}^+$ and $v_{K_i}^-$ agree there, and by Fact 2.1(a) (one-sided continuity) they are continuous there; so h_i is differentiable with the

continuous derivative $h'_i(t) = \langle v_{K_i}^+(t), v_t \rangle$, and ϱ_0, ϱ_1 are continuous. Hence $\varrho_0 = \varrho_1$ everywhere on (a, b) . We read this off for the two kinds of curves.

Lemma 8.2 (the tangent equations). *Let (a, b) be one of the four intervals.*

- (a) *Suppose that $\mathbf{z}_K = \mathbf{I}_K^T$ with $b \leq T < a + \pi$. Then there are $p, q \in \mathbb{R}$ with $\Delta(t) = p \cos t + q \sin t$ for $t \in [a, b]$, and $\Delta(T) = p \cos T + q \sin T$.*
- (b) *Suppose that $\mathbf{z}_K = \mathbf{y}_K$ (the second term, $(a, b) = (\varphi, \pi/2 - \varphi)$). Then $\Delta(t) = \Delta(b) - \int_t^b \Delta(u + \pi/2) du$ for $t \in [a, b]$.*

Proof. (a) For $t \in (a, b)$ we have $0 < T - t < \pi$, and $\langle \mathbf{I}_K^T(t), v_t \rangle = (h_K(T) - h_K(t) \cos(T - t)) / \sin(T - t)$, so $\varrho_i(t) = \frac{h_i(T) - h_i(t) \cos(T - t)}{\sin(T - t)} - h'_i(t)$. The equality $\varrho_0 = \varrho_1$ is

$$\sin(T - t)\Delta'(t) + \cos(T - t)\Delta(t) = \Delta(T).$$

Then $\chi(t) = (\Delta(t) - \Delta(T) \cos(T - t)) / \sin(T - t)$ has $\chi' = [\Delta' \sin(T - t) + \Delta \cos(T - t) - \Delta(T)] / \sin^2(T - t) = 0$ on (a, b) , so $\chi \equiv \kappa$ and $\Delta(t) = \Delta(T) \cos(T - t) + \kappa \sin(T - t) = p \cos t + q \sin t$, with $p = \Delta(T) \cos T + \kappa \sin T$ and $q = \Delta(T) \sin T - \kappa \cos T$; then $p \cos T + q \sin T = \Delta(T)$. By continuity this holds on $[a, b]$.

(b) Here $\langle \mathbf{y}_K(t) - v_K^+(t), v_t \rangle = h_K(t + \pi/2) - h'_K(t)$, so $\varrho_0 = \varrho_1$ says $\Delta'(t) = \Delta(t + \pi/2)$ on (a, b) . Integrate; Δ is continuous. $\square$

8.3. The cap is a horizontal translate.

Proposition 8.3 (the difference of support functions). $\Delta(t) = s \cos t$ for $t \in [0, \pi]$, where $s = -\Delta(\pi)$.

Proof. We solve the four equations in reverse order, using $\Delta(\pi/2) = 0$.

On $[\pi/2, \pi]$ (target $T = \pi$): $\Delta = p \cos t + q \sin t$ and $\Delta(\pi/2) = q = 0$, so $\Delta(t) = p \cos t$ and $\Delta(\pi) = -p$; thus $\Delta(t) = s \cos t$ on $[\pi/2, \pi]$ with $s = -\Delta(\pi)$.

On $[\pi/2 - \varphi, \pi/2]$ (target $T = \pi - \varphi$): $\Delta = p' \cos t + q' \sin t$, $\Delta(\pi/2) = q' = 0$, and $\Delta(\pi - \varphi) = p' \cos(\pi - \varphi)$. By the previous step $\Delta(\pi - \varphi) = s \cos(\pi - \varphi)$, and $\cos(\pi - \varphi) \neq 0$, so $p' = s$.

On $[\varphi, \pi/2 - \varphi]$: with $b = \pi/2 - \varphi$, $\Delta(u + \pi/2) = s \cos(u + \pi/2) = -s \sin u$ for $u \in [\varphi, b]$ by the first step, and $\Delta(b) = s \cos b$ by the second, so $\Delta(t) = s \cos b + s \int_t^b \sin u du = s \cos t$.

On $[0, \varphi]$ (target $T = \pi/2$): $\Delta = p'' \cos t + q'' \sin t$, $\Delta(\pi/2) = q'' = 0$, and $\Delta(\varphi) = s \cos \varphi = p'' \cos \varphi$ gives $p'' = s$. $\square$

Lemma 8.4 (caps with translated supports). *If $K, C \in \mathcal{K}_{\pi/2}^c$ and $h_K(t) - h_C(t) = s \cos t$ for $t \in [0, \pi]$, then $K = C + (s, 0)$ and $K \setminus \mathcal{N}(K) = (C \setminus \mathcal{N}(C)) + (s, 0)$.*

Proof. By (2.2) with $\omega = \pi/2$, $K = \{p : p_y \geq 0, \langle p, u_t \rangle \leq h_K(t) \text{ for } t \in [0, \pi]\}$, and likewise for C . Since $h_{C+(s,0)}(t) = h_C(t) + s \cos t$ and $(s, 0)$ is

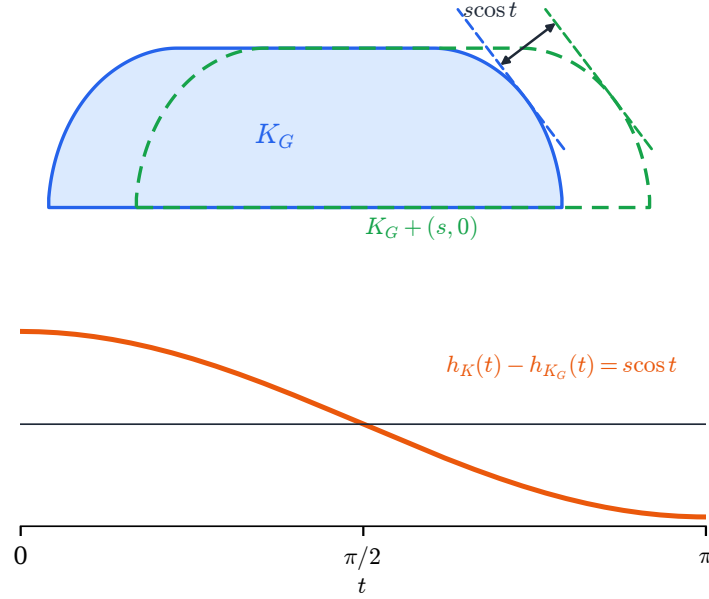


FIGURE 8. Proposition 8.3 and Lemma 8.4. Top: Gerver's cap K_G and its horizontal translate by $(s, 0)$; their supporting lines at the normal t lie $s \cos t$ apart. Bottom: the difference $h_K - h_{K_G} = s \cos t$ of the support functions on $[0, \pi]$.

parallel to the floor, $C + (s, 0)$ is the same set as K . Finally $Q_{C+w}^-(t) = Q_C^-(t) + w$ and $F_{\pi/2} + (s, 0) = F_{\pi/2}$. $\square$

Theorem 8.5 (maximizing right-angle caps). *A cap $K \in \mathcal{K}_{\pi/2}^c$ has $\mathcal{A}(K) = |G|$ if and only if $K = K_G + (s, 0)$ for some $s \in \mathbb{R}$. In this case $K \setminus \mathcal{N}(K) = G + (s, 0)$. Consequently the maximizers of $\mathcal{A}$ on $\mathcal{K}_{\pi/2}^c$ are the horizontal translates of Gerver's cap, and every monotone sofa U with rotation angle $\pi/2$ and $|U| = |G|$ is a horizontal translate of G .*

Proof. If $\mathcal{A}(K) = |G|$, then $K \in \mathcal{K}^i$ by Corollary 7.13; Proposition 8.3 and Lemma 8.4 (with $C = K_G$) give $K = K_G + (s, 0)$ and $K \setminus \mathcal{N}(K) = (K_G \setminus \mathcal{N}(K_G)) + (s, 0) = G + (s, 0)$ by Fact 2.16(b). Conversely $\mathcal{A}(K_G) = |G|$ by Fact 2.4(c), and $\mathcal{A}(C + (s, 0)) = \mathcal{A}(C)$ for every cap C , since $\mathcal{N}(C + (s, 0)) = \mathcal{N}(C) + (s, 0)$. The maximum of $\mathcal{A}$ is $|G|$ by Fact 2.7. If U is a monotone sofa with rotation angle $\pi/2$ and $|U| = |G|$, its cap $K = \mathcal{C}(U)$ has $\mathcal{A}(K) = |U| = |G|$ (Fact 2.4(c)), and $U = K \setminus \mathcal{N}(K)$. $\square$

9. RECOVERING THE SOFA; PROOF OF THEOREM 1.1

By (6.1) and Theorem 8.5, a rigid image of S lies in a horizontal translate of G and has the same area. A closed subset of a set of finite area, with the

same area, is the whole set if that set is the closure of its interior (Lemma 9.1), but not in general (Figure 9, Remark 9.2). So it remains to show that G is.

Lemma 9.1 (recovering a set from its area). *Let $X \subset Y \subset \mathbb{R}^2$ with X closed, $Y = \text{cl}(\text{int } Y)$, $|Y| < \infty$ and $|X| = |Y|$. Then $X = Y$.*

Proof. $|Y \setminus X| = 0$. The set $\text{int } Y \setminus X$ is open, and it has measure 0, so it is empty. Hence $\text{int } Y \subset X$, and since X is closed, $Y = \text{cl}(\text{int } Y) \subset X$. $\square$

Remark 9.2. Both hypotheses on Y are needed. The closed connected set $Y = [0, 1]^2 \cup ([1, 2] \times \{0\})$ and its closed subset $X = [0, 1]^2$ have the same finite area, and Y is not the closure of its interior. For $Y = \mathbb{R}^2$ and $X = \mathbb{R}^2$ minus the open unit disc, $Y = \text{cl}(\text{int } Y)$ and $|X| = |Y| = \infty$. For $Y = G$, a closed subset that misses an interior point of G misses a disc about it, and so loses area.

Lemma 9.3 (removing the region under an envelope). *Let $K \subset \mathbb{R}^2$ be closed with $K = \text{cl}(\text{int } K)$ and $[a, b] \times [0, 1] \subset K$. Let Γ be a compact set of points with abscissa in $[a, b]$ and height in $[0, 1]$, and let $\Gamma_{<}$ be the set of points q with $q_y \geq 0$ that lie strictly below a point γ of Γ with $\gamma_x = q_x$. If $X = K \setminus \Gamma_{<}$ is closed, then $X = \text{cl}(\text{int } X)$.*

Proof. Let $\bar{\Gamma}$ be the set of points q with $q_y \geq 0$ that lie on or below a point of Γ with the same abscissa; it is the image of the compact set $\Gamma \times [0, 1]$ under $(\gamma, \lambda) \mapsto (\gamma_x, \lambda\gamma_y)$, hence compact, and $\Gamma_{<} \subset \bar{\Gamma}$. Let $p \in X$.

If $p \notin \bar{\Gamma}$, then p has a neighborhood outside $\bar{\Gamma}$, and since $K = \text{cl}(\text{int } K)$ it is a limit of interior points of K outside $\bar{\Gamma}$; these are interior points of $X = K \setminus \Gamma_{<}$. So $p \in \text{cl}(\text{int } X)$.

If $p \in \bar{\Gamma}$, then $p \notin \Gamma_{<}$ means that p is the highest point of Γ on its vertical line: $p_x \in [a, b]$ and $p_y < 1$. The points $(p_x, p_y + s)$, $0 < s \leq 1 - p_y$, lie in the rectangle $[a, b] \times [0, 1] \subset K$ and outside $\bar{\Gamma}$, so by the first case they lie in $\text{cl}(\text{int } X)$, and so does their limit p .

So $X \subset \text{cl}(\text{int } X)$, and $\text{cl}(\text{int } X) \subset X$ since X is closed. $\square$

Proposition 9.4 (Gerver's sofa is regular closed). $G = \text{cl}(\text{int } G)$.

Proof. Let $\Gamma = \mathbf{D}([t_0, t_2]) \cup \mathbf{x}([t_1, t_4]) \cup \mathbf{B}([t_3, t_5])$, $a = \mathbf{D}(0)_x$ and $b = \mathbf{B}(\pi/2)_x$. By Fact 2.16(b) and (c), $G = K_G \setminus \mathcal{N}(K_G) = K_G \setminus \Gamma_{<}$, which is closed, being a moving sofa. The set Γ is compact. The abscissa increases along $\mathbf{D}$ and $\mathbf{B}$ and, as t grows, decreases along $\mathbf{x}$, and the endpoints are joined, so

$$a = \mathbf{D}(0)_x < \mathbf{D}(t_2)_x = \mathbf{x}(t_4)_x < \mathbf{x}(t_1)_x = \mathbf{B}(t_3)_x < \mathbf{B}(t_5)_x = b,$$

and the abscissa of every point of Γ lies in $[a, b]$. The ordinate of $\mathbf{D}$ is nondecreasing, since $\mathbf{D}'$ is a positive multiple of u_t and $\sin t \geq 0$; it starts at 0 and ends at $\mathbf{x}(t_4)_y < 1$. The ordinate of $\mathbf{B}$ is nonincreasing, since $\mathbf{B}'$ is a negative multiple of v_t ; it ends at 0 and starts at $\mathbf{x}(t_1)_y < 1$. The ordinate of $\mathbf{x}$ on $[t_1, t_4]$ lies in $(0, 1)$ by Fact 2.17. So the heights of Γ lie in $[0, 1]$.

The points $(a, 1) = \mathbf{D}(0) + v_0 = \mathbf{C}(0)$ and $(b, 1) = \mathbf{B}(\pi/2) + u_{\pi/2} = \mathbf{A}(\pi/2)$ are vertices of the cap K_G (Fact 2.16(b)), so they lie in K_G . A cap with

rotation angle $\pi/2$ is closed downward above the floor: for $p \in K$ and $0 \leq \lambda \leq p_y$ we have $p - (0, \lambda) \in K$, because $\langle (0, -\lambda), u_s \rangle \leq 0$ for $s \in [0, \pi]$. Hence $K_G \supset [a, b] \times [0, 1]$, so K_G is a closed convex set with nonempty interior, and $K_G = \text{cl}(\text{int } K_G)$. Lemma 9.3 applies with $K = K_G$. $\square$

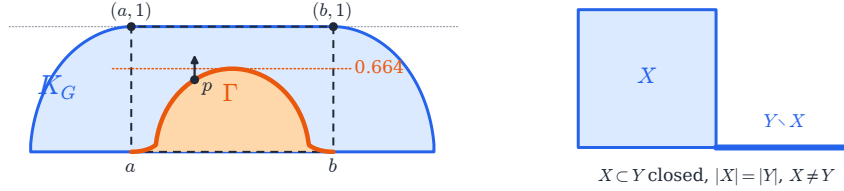


FIGURE 9. Proposition 9.4. Left: Gerver's cap K_G with its niche (orange) below the curve Γ formed by $\mathbf{D}$, $\mathbf{x}$ and $\mathbf{B}$; the dashed rectangle $[a, b] \times [0, 1]$ lies in K_G , and the greatest height of Γ is $0.664\dots$. Every point p of Γ is a limit of points of G above it. Right: why $\text{cl}(\text{int } Y) = Y$ is needed in Lemma 9.1: a closed subset of Y with the same area misses no interior point.

Proof of Theorem 1.1. Let S be a moving sofa with $|S| = |G|$. By Proposition 3.2, Proposition 6.9 and Theorem 8.5, there are $\omega \in [\omega_0, \pi/2]$, $\psi = \pi/2 - \omega$, $w_0, w_1 \in \mathbb{R}^2$ and $s \in \mathbb{R}$ such that

$$g_1(S) \subset U = G + (s, 0), \quad g_1(p) = R_\psi(p + w_0) + w_1.$$

Let $g(p) = g_1(p) - (s, 0) = R_\psi p + v$ with $v = R_\psi w_0 + w_1 - (s, 0)$. Then $g(S) \subset G$. The set $g(S)$ is closed, since S is closed and g is a homeomorphism, and $|g(S)| = |S| = |G| < \infty$ since g preserves area. By Proposition 9.4, $G = \text{cl}(\text{int } G)$, so $g(S) = G$ by Lemma 9.1, that is, $R_\psi S + v = G$. This is the assertion with $\theta = \psi$. $\square$

The proof also gives the following description of all maximal sofas.

Corollary 9.5 (the maximal sofas). *A set S is a moving sofa of maximal area if and only if it is a moving sofa and $R_\theta S + v = G$ for some θ and v .*

Proof. One direction is Theorem 1.1 together with $|S| = |G|$ being the maximum (Fact 2.7). Conversely a rigid image of G has the area of G . $\square$

Remark 9.6 (no rotation is needed). In Theorem 1.1 one can take $\theta = 0$: every moving sofa S with $|S| = |G|$ is a translate of G , and so has the rotation angle $\pi/2$. This refinement is not part of the formal statement. Indeed, a moving sofa lies after a translation in the strip $\mathbb{R} \times [0, 1]$, so $|\langle p - q, u_{\pi/2} \rangle| \leq 1$ for $p, q \in S$. In the proof above $\theta = \psi \in [0, \pi/2 - \omega_0]$ and $S = R_{-\psi}(G - v)$, so the width of G in the direction $u_{\pi/2+\psi}$ is at most one. But the width of G in the

direction u_r exceeds one for every $r \in [0, \pi] \setminus \{\pi/2\}$, and $\pi/2 + \psi \in [\pi/2, \pi]$; so $\psi = 0$.

To see the claim about the width, let $a < b$ be as in Proposition 9.4, and let $x_- = \mathbf{C}(\pi/2)_x$. The points $(a, 1) = \mathbf{C}(0)$ and $(b, 1) = \mathbf{A}(\pi/2)$ (the ends of the top edge of K_G), $(1, 0) = \mathbf{A}(0)$ (Appendix A, condition (2)) and $(x_-, 0) = \mathbf{C}(\pi/2)$ (the ends of the floor, by (2.3), Fact 2.13(d) and Fact 2.16(b)) lie in K_G , and not in $\mathcal{N}(K_G)$: the points of $\mathcal{N}(K_G)$ have height less than 1 and abscissa in (a, b) (the only points of Γ with abscissa a or b are $\mathbf{D}(t_0)$ and $\mathbf{B}(t_5)$, of height 0), while $x_- = -h_{K_G}(\pi) \leq a < b \leq h_{K_G}(0) = 1$. So they lie in G . For $r \in [0, \pi/2]$ the width of G in the direction u_r is at least $\langle (b, 1) - (x_-, 0), u_r \rangle = (b - x_-) \cos r + \sin r > \cos r + \sin r \geq 1$, because $b - x_- > 1$: indeed $x_- = \mathbf{C}(\pi/2)_x = \mathbf{x}(\pi/2)_x - 1$, and $b = \mathbf{B}(t_5)_x > \mathbf{B}(t_3)_x = \mathbf{x}(t_1)_x > \mathbf{x}(t_4)_x \geq \mathbf{x}(\pi/2)_x$ by Fact 2.16(c). For $r \in (\pi/2, \pi]$ it is at least $\langle (a, 1) - (1, 0), u_r \rangle = (1 - a)|\cos r| + \sin r > |\cos r| + \sin r \geq 1$, because $a < 0$: $a = \mathbf{D}(t_0)_x < \mathbf{D}(t_2)_x = \mathbf{x}(t_4)_x < \mathbf{x}(t_1)_x \leq \mathbf{x}(0)_x = 0$ by Fact 2.16(c) and Appendix A, condition (2). (Numerically, $a = -1.4206\dots$, $b = 0.1931\dots$ and $x_- = -2.2275\dots$)

10. FORMALIZATION, PROVENANCE AND OPEN QUESTIONS

10.1. The formalization. The repository [14] (Apache-2.0) contains three Lean 4 libraries on top of Mathlib (Lean and Mathlib v4.35.0-rc3):

- **MovingSofaOptimality** (51 files, about 40,000 lines): Baek's paper [1], one directory per chapter, with the results that it cites (Schneider's area formula for planar convex bodies; the existence and uniqueness of the solution of Romik's system in the box of Fact 2.16) and the structure of Gerver's sofa (Fact 2.16(b),(c)), which [1] states without proof (Thm. 8.4.1) or proves invalidly (Thm. 6.1.2);
- **MovingSofaUniqueness** (8 files, about 5,000 lines): Sections 4 to 9, one module per step of the argument;
- **MovingSofaBridge** (4 files, about 2,600 lines): the connection with the definitions of formal-conjectures [4], which are different (a motion there is a path of isometries that starts at the identity, and Gerver's sofa is defined from Gerver's four constants); it shows that a set is a moving sofa for formal-conjectures if and only if it lies in the horizontal side of the hallway and is a moving sofa in the sense of Appendix C, and that the optimal areas and the Gerver's sofas of the two formulations agree, so that formal-conjectures' statements, the open uniqueness statement among them, follow from Theorem 1.1.

The file `Challenge.lean` states twelve theorems, among them Baek's theorem (`Baek.gerver_sofa_optimal`) and Theorem 1.1 (`Baek.gerver_sofa_unique`). The five theorems of the namespace `Baek` use only Mathlib's vocabulary and the definitions of Appendices A and C; the other seven use the definitions of formal-conjectures, restated verbatim (one topology instance is renamed), and the three theorems of the namespace `Bridge`

also use Baek's definitions. `Solution.lean` proves them: the Challenge is a file of statements, so its twelve theorems end in `sorry` by design, and Comparator checks that `Solution.lean` proves exactly the statements of `Challenge.lean`. `lake build` succeeds, and an audit checks that every declaration of the three libraries depends only on the axioms `propext`, `Classical.choice` and `Quot.sound`. The project is registered in the Palomar registry (entry PALOMAR-2026-10-02-000008), and the continuous integration and Palomar's preflight both passed on the commit cited in [14]. To rebuild: `lake exe cache get && lake build`, then the commands listed in the repository's `README.md`.

What the machine check gives is this: Theorem 1.1, *as the Lean statement says it*, follows from Lean's axioms and Mathlib, with Baek's argument and the argument of this paper both included. It does not check that the definitions say what they should; Appendix C reproduces the Lean definitions (moving sofa, hallway, the rotation path and Gerver's sofa) in mathematical notation for that purpose. Every proof of a result of [1] in the library follows Baek's argument, except at sixteen steps, each forced by an error or gap of the paper (E12, E15, E17, E20, E21, E24), by mathematics that Mathlib lacks (the Jordan curve theorem, Green's theorem, the Brunn–Minkowski inequality, mixed volumes), or by the representations chosen in the formalization (the surface area measure as a Lebesgue–Stieltjes measure, the area functional of a convex arc as $\frac{1}{2} \int h d\sigma$, no orientations of curves). A route check in continuous integration compares the results used by each Lean proof with those cited by Baek's proof; the other details were compared with Baek's TeX source by AI agents, not by a person.

The text of Sections 4 to 9 follows the formal proofs, but it is a translation of them, and it is not machine-checked. It differs from the Lean in the following places. The existence of penalized maximizers (Proposition 4.5) takes a Hausdorff limit and shows that it is a polygon cap, whereas the Lean proof takes a subsequence along which the finitely many support values converge and forms the polygon cap from the limit values. Some statements are slightly sharper than their Lean versions: the box of Lemma 4.4 is $[-R, R] \times [0, R]$ where Lean has $[-R, R] \times [0, 2R]$, the constant of Lemma 7.5 has B where Lean has $2B$, and Lemma 6.6 puts the vertices in the open quarter-plane, which Lean proves with other declarations than the closed version. Of Fact 2.10(c) only the sine form is formalized. Lemma 7.4 is stated in Lean in Baek's form, with $g_K^-(t + \delta)$, Lemma 7.6 is inlined in one declaration, and the Lean proof of Lemma 2.5 derives $O \in K$ from (2.3), not from $o_\omega \in K$. The converse part of Theorem 8.5 and Remark 9.6 are proved only in the text, and Corollary 9.5 is stated in Lean only in the vocabulary of formal-conjectures, with isometries in place of rotations and translations.

The informal argument that preceded the formalization [14, note 20] differs from both in the following places: the penalty is fixed and sampled at persistent dyadic normals instead of vanishing and continuous, and there is no box constraint at the right angle (Section 4); the selection needs $\mathcal{A}_\omega(K_*) > 0$;

the arms are compared with Baek’s lower sequence instead of by a maximum-deficit argument (Section 7); a cap is described by its upper support values and the floor (Section 8); and the regular closedness of Gerver’s sofa uses that its rotation path stays strictly below the top of the cap, which is proved by interval arithmetic (Fact 2.17).

10.2. Provenance of the argument, and the use of AI. The informal argument was written by ChatGPT Pro 6 (OpenAI) on 2 October 2026, at the request of the author, who asked for a proof in Baek’s definitions that every moving sofa of maximal area is congruent to Gerver’s; it came with a draft of the formalization and a draft of the connection with formal-conjectures that had never been compiled. The formalization of [1] and the completion of the drafts, in which 54 proofs of the first and 78 proofs of the second that did not compile were replaced and proved again, were done by Claude Opus 5.5 (Anthropic) in Claude Code, with sub-agents, following the author’s **formalize-math-paper** procedure (rounds between 1 and 3 October 2026); [14] records who did what, with times and sizes. This manuscript was written by Claude Sonnet 5.5 (Anthropic) in Claude Code on 4 October 2026, at the author’s request, from the Lean library, the illustrated text of the proofs in [14] and the text of [1]; the figures were computed from the definitions by the script `docs/paper/figures/make_figures.py` of the repository, which was added after the commit cited in [14]. The author asked for each of these steps and decided its scope. None of the three systems is an author of this paper: they cannot take responsibility for it.

The text of this paper is not machine-checked, and the correspondence between the text and the Lean statements is given by the dictionary of Appendix C. On 4 October 2026, six runs of Claude Opus 5.5, each assigned some sections, compared the statements and proofs of this paper with the Lean statements and with the TeX source of [1]; a run of Claude Sonnet 5.5 checked the claims about the formalization, the dictionary and the references; and a run of Claude Opus 5.5 read the whole paper without access to either. The findings were corrected. This is not a refereeing: at the time of writing, the argument has not been refereed, and no person has checked it independently of the formalization.

10.3. Open questions. *Stability.* Is there a quantitative version of Theorem 1.1: a bound for the distance of a moving sofa S to the nearest copy of G in terms of $|G| - |S|$? The obstacle is that Baek’s bound $\mathcal{A} \leq \mathcal{Q}$ is available only for caps with the injectivity condition, which we establish for sofas of maximal area, through limits that carry no rate.

Other corners. For a hallway whose corner has an angle $\alpha \neq \pi/2$ one can ask for the largest area as a function of α and for the extremal shapes. Each step of Baek’s method has an analogue to prove, and its numerical inequalities, such as those of Fact 6.5, are specific to the right angle.

The ambidextrous sofa. Romik [10] found a sofa of area 1.6449... that can turn both left and right around a right-angled corner. As far as we know,

it is not known whether it is optimal. Baek's reduction to monotone sofas uses one hallway; for a sofa that must turn both ways, the cap, the niche and the bound $\mathcal{Q}$ all need two-sided analogues.

Acknowledgments. The results of [1] are the foundation of this paper. The audit of its statements owes twelve findings to the notes of another formalization of it [9], and was also compared with the blueprint of a third [12].

APPENDIX A. GERVER'S SOFA

We recall Romik's description of Gerver's sofa [10, Section 4], in the form used by [1] and by the formalization. Let $t_0 = 0 < t_1 = \varphi < t_2 = \theta < t_3 = \pi/2 - \theta < t_4 = \pi/2 - \varphi < t_5 = \pi/2$. A *parameter tuple* consists of the angles φ, θ , numbers $a_1, a_2, b_1, b_2, c_1, c_2, d_1, d_2, e_1, e_2$ and points $\kappa_1, \dots, \kappa_5 \in \mathbb{R}^2$ (22 real parameters). The *rotation path* $\mathbf{x} : [0, \pi/2] \rightarrow \mathbb{R}^2$ is $\mathbf{x} = \mathbf{x}_i$ on the i th phase, where the phases are $[0, t_1)$, $[t_1, t_2)$, $[t_2, t_3]$, $(t_3, t_4]$ and $(t_4, t_5]$ (for a solution the pieces agree at the ends of the phases, by condition (3) below, so the conventions at the ends matter only for other tuples), and

$$\begin{aligned} \mathbf{x}_1(t) &= R_t(a_1 \cos t + a_2 \sin t - 1, -a_2 \cos t + a_1 \sin t - \tfrac{1}{2}) + \kappa_1, \\ \mathbf{x}_2(t) &= R_t(-\tfrac{t^2}{4} + b_1 t + b_2, \tfrac{t}{2} - b_1 - 1) + \kappa_2, \\ \mathbf{x}_3(t) &= R_t(c_1 - t, c_2 + t) + \kappa_3, \\ \mathbf{x}_4(t) &= R_t(-\tfrac{t}{2} + d_1 - 1, -\tfrac{t^2}{4} + d_1 t + d_2) + \kappa_4, \\ \mathbf{x}_5(t) &= R_t(e_1 \cos t + e_2 \sin t - \tfrac{1}{2}, -e_2 \cos t + e_1 \sin t - 1) + \kappa_5. \end{aligned}$$

For a differentiable curve $\mathbf{x}$ let $\alpha = \langle \mathbf{x}', u_t \rangle$, $\beta = \langle \mathbf{x}', v_t \rangle$, and let $\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}$ be as in Section 2.6. The tuple *solves Romik's equations* if $0 < \varphi < \theta < \pi/4$ and:

- (1) (left-right symmetry) $e_1 = a_1$, $e_2 = -a_2$, $d_1 = \frac{\pi}{4} - b_1$, $d_2 = b_2 + \frac{\pi}{4}(2b_1 - \frac{\pi}{4})$, $c_2 = c_1 - \frac{\pi}{2}$;
- (2) (the start) $\kappa_1 = (1 - a_1, \frac{1}{4})$ and $a_2 = -\frac{1}{4}$, so that $\mathbf{x}(0) = O$ and $\mathbf{A}(0) = (1, 0)$;
- (3) (continuous differentiability) $\mathbf{x}_i(t_i) = \mathbf{x}_{i+1}(t_i)$ and $\mathbf{x}'_i(t_i) = \mathbf{x}'_{i+1}(t_i)$ for $i = 1, 2, 3, 4$;
- (4) (the contact conditions) $\mathbf{x}_1(t_1) = \mathbf{B}_4(t_3)$ and $\mathbf{x}_5(t_4) = \mathbf{D}_2(t_2)$, where $\mathbf{B}_4 = \mathbf{x}_4 + \alpha_4 v_t$ and $\mathbf{D}_2 = \mathbf{x}_2 - \beta_2 u_t$ are the contact curves of $\mathbf{x}_4$ and $\mathbf{x}_2$.

These are Romik's equations (27)–(44). *Gerver's sofa* is the shape of the rotation path of the solution with $\varphi \in [0.039, 0.04]$ and $\theta \in [0.68, 0.69]$, which is unique in this box (Fact 2.16(a)). Its numerical values, to ten decimal places (computed outside the formalization, which uses rigorous enclosures),

are

$$\begin{aligned}
\varphi &= 0.0391773648, & \theta &= 0.6813015094, \\
a_1 = e_1 &= 1.2103224221, & a_2 = -e_2 &= -0.25, \\
b_1 &= -0.5276245980, & b_2 &= 0.9202583852, \\
c_1 &= 0.6260455228, & c_2 &= -0.9447508039, \\
d_1 &= 1.3130227614, & d_2 &= -0.5253826704, \\
\kappa_1 &= (-0.2103224221, 0.25), & \kappa_5 &= (-1.0172040368, 0.25), \\
\kappa_2 &= (-0.9191792928, 0.4724066198), \\
\kappa_3 &= (-0.6137632294, 0.8896264790), \\
\kappa_4 &= (-0.3083471661, 0.4724066198),
\end{aligned}$$

and $\mathbf{x}(\pi/2) = (-1.2275264589, 0)$, $\max_t \mathbf{x}(t)_y = \mathbf{x}(\pi/4)_y = 0.6642678135$, $\mathbf{x}(t_1)_y = \mathbf{x}(t_4)_y = 0.0551867605$. The area is $|G| = 2.21953166887\dots$ (Gerver, Romik), and $2.2192 \leq |G| \leq 2.2199$ is proved in [14]. In Gerver's own parametrization the constants $A = 0.0944265608$ and $B = 1.3992037273$ together with φ and θ solve a system of four equations [3], from which the 22 parameters above are computed.

The contact curves **A**, **B**, **C**, **D** are the points where the sides of the hallway touch the sofa. The sofa touches the walls $a(t)$ and $c(t)$ at **A**(t) and **C**(t) for all t ; the wall $d(t)$ at **D**(t) for $t \in [t_0, t_2]$; the wall $b(t)$ at **B**(t) for $t \in [t_3, t_5]$; and the inner corner touches it at $\mathbf{x}(t)$ for $t \in [t_1, t_4]$. The boundary of G consists of 18 pieces: **A** on four phases, the top edge, **C** on four phases, an edge on the x -axis, **D** on two phases, $\mathbf{x}$ on three phases, **B** on two phases, and a second edge on the x -axis.

APPENDIX B. CORRECTIONS TO THE STATEMENTS OF [1] THAT ARE USED HERE

The audit that accompanies the formalization compared every result of [1] with its TeX source and with the formal statement [14]. Every result of the paper survives with the intended statement. Table 1 lists the corrections that concern the results used in this paper, with the labels En of the audit (its complete list has 27 items, E22 only pointing to E21, and two missing hypotheses); the *missing hypotheses* are the following. [1, Thm. 2.1.1] (Schneider's Theorem 4.2.3: $\sigma_K(X)$ is the length of $\bigcup_{t \in X} e_K(t)$) needs K to have interior points, and [1, Thm. 3.1.2] needs a bounded Nef polygon; every use satisfies them (the text itself uses σ_K only as defined in Fact 2.1(b), which does not depend on Thm. 2.1.1). Twelve of the findings in the audit are due to the notes of the formalization [9].

Table 1: Corrections to [1] that concern the results used here.

Item	Result of [1]	Correction	Where here
E1	Def. 2.3.9	the left and right side of a line are defined for lines not parallel to the x -axis (printed: y -axis)	Fact 2.4(a)
E2	Thm. 2.4.1, 2.4.2	the union over $t \in [0, \omega]$ and over $(0, \omega)$ agree, since $F_\omega \cap Q_K^-(0) = F_\omega \cap Q_K^-(\omega) = \emptyset$; $\mathcal{C}(S)$ is bounded also for $\omega = \pi/2$	Fact 2.4(b)
E3	Prop. 2.5.4	the reflection exchanges the walls ($a \leftrightarrow c$, $b \leftrightarrow d$) and the ends $W \leftrightarrow Z$; K and not K^m stands on the right of the vertex identity	Fact 2.4(d), Lemma 6.6, Proposition 7.9
E4	Lemma 2.5.6, Thm. 2.5.8 (and Lemma 3.4.8, Remark 3.4.1)	used without proof, true for every cap: $A_K^-(0) = (h_K(0), 0)$, $C_K^+(\omega) = h_K(\omega + \pi/2)v_\omega$, and $O \in K$ if $\omega < \pi/2$; the same holds for o_ω , which Lemma 3.4.8 and Remark 3.4.1 use (not in the audit)	Lemma 2.6, Lemma 2.5, Lemma 6.1, Lemma 5.3
E5	Thm. 3.1.2, Lemma 3.4.7	the Nef polygon must be bounded; the two applications of the theorem for a pinned normal change disjoint strips	Fact 2.11
E6	Def. 3.2.5	$\mathcal{N}_\Theta(K) = F_\omega \cap \bigcup_{t \in \Theta} Q_K^-(t)$ (printed: P_ω in place of F_ω); with P_ω , Prop. 3.3.5 and Lemma 3.4.2 are false	Section 2.3, Fact 2.9, Proposition 4.5
E7	Prop. 3.3.1 (2)	normal angles in $\Theta^\circ \cup \{\omega + \pi, 3\pi/2\}$ (printed: Θ°)	Section 2.3
E8	Thm. 3.4.3, 3.5.2, 3.5.4	minor gaps: a Hausdorff limit of polygon caps is a polygon cap (resp. a cap), and the niche of the limit lies in the polygon niches eventually, point by point	Proposition 4.5, Proposition 4.8, Lemma 4.6
E9	Lemma 4.2.4	the range is $\omega \in [\sec^{-1}(2.2), \pi/2)$ (printed: $[\tan^{-1}(2.2), \pi/2)$)	Fact 6.5

Item	Result of [1]	Correction (continued)	Where here
E10	Thm. 4.2.5, 1.5.2	the width bound for the pentagon $P_\omega \setminus \Delta_\omega$ is asserted without proof; the proof puts the vertices of Δ_ω in the open quarter-plane	Lemma 6.6, Lemma 6.7
E12	Thm. 6.1.2	the proof (Gerver's Thm. 2 makes G a balanced maximum sofa) is invalid; the statement is true and follows from Romik's equations	Fact 2.16(b)
E13	Prop. 6.2.2	$f_{K^m}^\pm(t) = g_K^\mp(\pi/2 - t)$, and likewise for g (printed: without the change of angle)	Fact 2.13(c)
E14	Def. 6.2.2, Thm. 6.2.3	∂^+ is the right derivative (Def. 6.2.2 calls it the left one), and the second display of Thm. 6.2.3 is about $\partial^- \mathbf{x}_K$	Fact 2.13(a)
E15	Lemma 6.3.1	cites Thm. 3.5.4 (balanced maximum caps) for $\mathcal{N}(K) \subset K$ of a maximum polygon cap; the applicable result is Thm. 3.4.10, which gives what the lemma needs, and the lemma holds	not used here (the box of Proposition 4.5 bounds the diameter)
E17	Lemma 6.5.2– 6.5.5, Thm. 6.5.6	$f_K(0) = 1$ is used without being stated; Lemma 6.5.5 is used on $(0, \pi/2]$ (printed: $(0, 1]$)	Fact 2.13(d), Fact 2.14(b)
E20	Lemma 8.3.5	the arc $\mathbf{u}_K^{0,\pi}$ is outside the range $b < a + \pi$ of [1, Thm. 7.3.2]; the area is split at the atom $\pi/2$ instead	Fact 2.15(b)
E21	Lemma 8.1.7, Thm. 8.2.4	$\mathcal{N}(K) \subset K$ is used but not available on $\mathcal{K}^1$; both statements hold and have proofs that avoid it	Fact 2.15(a)
E23	Lemma 8.3.6 (3)	the tangent line parametrization is evaluated at $\pi/2$ (printed: at φ^L , where the statement is false)	Fact 2.15(b)

Item	Result of [1]	Correction (continued)	Where here
E24	Thm. 8.4.1	stated without proof; true, and proved from Romik's equations by interval arithmetic; part (4) holds with one-sided derivatives at the junctions and the ends of the phases. Not in the audit: part (2), read as a Jordan region, omits the open floor segment of $\mathcal{N}(K_G)$, and the exact description is the identity in Fact 2.16(c)	Fact 2.16
E25, E26	Thm. 8.4.3 (2), Prop. 8.4.4 (4)	$\mathbf{B}(t_3)$ (printed: $\mathbf{D}(t_3)$); the measure $\check{\sigma}_D$ lives on $(\pi/2 + t_0, \pi/2 + t_2]$ with a shifted density	Fact 2.15(c)
E27	several	misprints: the second half-plane of $Q_S^-(t)$ lacks its “−1” (Prop. 2.2.2); the sign of the change of $ \mathcal{N}_\Theta $ in the proof of Lemma 3.4.7; $\mathcal{R}_B$ starts at $\pi + \varphi^R$, not $\pi/2 + \varphi^R$ (Def. 8.3.3); the statement of Thm. 2.1.3	Section 2.2, Fact 2.1, Fact 2.11, Section 2.5

APPENDIX C. THE LEAN FORMALIZATION: THE STATEMENT AND A DICTIONARY

C.1. The statement of Theorem 1.1. The statement of record is `Baek.gerver_sofa_unique` in `Challenge.lean`. It uses Mathlib's vocabulary and the definitions below, which `Challenge.lean` copies verbatim from `ChallengeDefs.lean`; they are those of this paper. In Lean the plane is $\mathbb{R} \times \mathbb{R}$, the area of a set is its Lebesgue measure `volume`, and `rot t` is the counterclockwise rotation by t .

- `horizSide` = $\{p : p_1 \leq 1, 0 \leq p_2 \leq 1\} = H_L$, `vertSide` = $\{p : 0 \leq p_1 \leq 1, p_2 \leq 1\} = V_L$, `hallway` = $H_L \cup V_L = L$.
- `IsMovingSofa S` holds if S is closed and connected (hence nonempty) and there are $\theta : \mathbb{R} \rightarrow \mathbb{R}$ and $c : \mathbb{R} \rightarrow \mathbb{R}^2$, continuous on $[0, 1]$, with $\theta(0) = 0$, $\text{rot}_{\theta(0)}p + c(0) \in H_L$ for $p \in S$, $\text{rot}_{\theta(s)}p + c(s) \in L$ for $p \in S$ and $s \in [0, 1]$, and $\text{rot}_{\theta(1)}p + c(1) \in V_L$ for $p \in S$ (Definition 2.2, the rotation angle $-\theta(1)$ being left free).
- `shapeOfPath x` = $H_L \cap \bigcap_{t \in [0, \pi/2]} (\mathbf{x}(t) + R_t L) \cap (\mathbf{x}(\pi/2) + R_{\pi/2} V_L)$, written in Lean with images of the hallway sets under $p \mapsto \mathbf{x}(t) + \text{rot}_t p$.

- **GerverParams** holds the 22 parameters of Appendix A; **P.path** is the rotation path glued from the five curves of Appendix A; **P.IsSolution** is the conjunction of $0 < \varphi < \theta < \pi/4$ and the four groups (1)–(4) of Appendix A (Romik’s equations (27)–(44)); **P.InBox** says $\varphi \in [0.039, 0.04]$, $\theta \in [0.68, 0.69]$; and **gerverSofa P = shapeOfPath P.path**.

The theorem: for every **P** : **GerverParams** with **P.IsSolution** and **P.InBox**, and every set $S \subset \mathbb{R} \times \mathbb{R}$ with **IsMovingSofa S** and **volume S = volume (gerverSofa P)**, there are $\theta \in \mathbb{R}$ and $v \in \mathbb{R} \times \mathbb{R}$ such that the image $\{\text{rot}_{\theta} p + v : p \in S\}$ equals **gerverSofa P**. The theorems **Baek.gerver_params_exists** and **Baek.gerver_params_unique** say that the box contains exactly one solution, so that **gerverSofa P** is Gerver’s sofa G , and **Baek.gerver_sofa_area** says that its area lies between 2.2192 and 2.2199.

C.2. Dictionary. Table 2 lists the declarations that prove the results of this paper. The modules are in the repository [14]: a name followed by a module in parentheses is in **MovingSofaUniqueness/** (**Main**, **Selection**, **Variation**, **AngleExtension**, **Curvature**, **Rigidity**, **RegularClosed**, **Rigid**) or, with a directory, in **MovingSofaOptimality/**. The results of [1] are in **MovingSofaOptimality/**, under the name **theorem $n_1.n_2.n_3$** (also **lemma**, **proposition**, **corollary**) for the result number $n_1.n_2.n_3$ of [1]. Some statements are slightly different in Lean: Lemma 4.4 has the box $[-R, R] \times [0, 2R]$, Lemma 7.5 has $2B$ for B , Lemma 6.6 has the closed quarter-plane in **consumed_of_pinned** (the open one is in the other declarations listed), Lemma 7.4 is stated with $g_K^-(t + \delta)$, and Fact 2.10(c) is formalized in its sine form (see Section 10 for the others). The converse part of Theorem 8.5 (a translate of K_G has sofa area $|G|$) and Remark 9.6 are not stated in Lean, and Corollary 9.5 only in the vocabulary of formal-conjectures (**FormalConjectures.MovingSofa.volume_eq_sofaConstant_iff_congruent_gerversSofa**).

Table 2: The results of this paper and the Lean declarations that prove them.

Result	Lean declarations
Theorem 1.1	Baek.gerver_sofa_unique (Challenge.lean); MovingSofaUniqueness.image_eq_gerver_of_volume_eq, maximizer_contained_in_gerver (Main)
Proposition 3.2	maximal_envelope , own_cap_isMax , isMaxCap_of_area_eq, cap_area_le_gerver (Main); Fact 2.7: theorem1_1_1, theorem3_5_5 , theorem3_5_6
Lemma 4.2, Lemma 4.3, Lemma 4.4	SupportSamples.penalty_recovery_zero, dyadic_recovery_ge , polygon_subset_sampleBox, niche_subset_box , polyNiche_subset_box (Selection)
Proposition 4.5	exists_penalizedMax , exists_dyadic_penalizedMax, selected_objective_ge , selected_sequence_bounded (Selection)

Result	Lean declarations (continued)
Lemma 4.6, Lemma 4.7, Proposition 4.8 Lemma 5.1, Proposition 5.2	dyadic_objective_limsup, caps_eq_of_dyadic_supports, exists_selectedCapSequence, penalty_tendsto_zero_of_objective (Selection) floatingCap, floatingCap_polygon, subset_floatingCap, floatingCap_support_bounds, floatingCap_support_other, floating_penalty_growth, floating_defect_le (Variation)
Lemma 5.3, Lemma 5.4, Proposition 5.5	pinned_contract_mem, pinned_div_mem, pinned_raw_support_bound, translated_strip_support, pinned_translation_bound, pinned_normalized_support_bound, pinned_defect_le (Variation)
Corollary 5.6	polygon_weighted_defect_zero, abs_selected_defect_le, selectorDefectBound (Variation)
Lemma 6.1, Lemma 6.2, Lemma 6.3, Proposition 6.4 Lemma 6.6, Fact 6.5, Lemma 6.7, Lemma 6.8, Proposition 6.9	ang_wedgeGapWInf_le_tau, ang_wedgeGapZInf_le_tau, lemma4_1_1, ang_abs_wedgeGapZInf_sub_le, ang_le_sigmaAt_of_tendsto (Angle/HorizontalSide); pinned_bounds_of_maximal_positive (Variation) consumed_of_pinned, right_angle_motion_of_pinned_bounds (AngleExtension); lemma4_2_2, lemma4_2_4, theorem4_2_5, ang_three_points_mem, ang_consumed_of_supp_zero, ang_mirror_mem_qMinus, ang_mem_niche_of_small, ang_width_le_one, right_angle_motion_of_width (Angle/RightAngle); maximal_monotone_has_right_angle (Main)
Lemma 7.1, Lemma 7.2, Lemma 7.3	polygon_tau_le_geom, polygon_curvature_with_defect (Curvature); inj_param_minus, inj_param_plus, inj_polygon_sigmaAt_eq, inj_not_mem_qMinus (Injectivity/DiscreteIneq)
Lemma 7.4, Lemma 7.5	polygon_arm_cell, polygon_step_with_error, polygon_Ico_with_errors, polygon_Ioo_with_errors (Curvature)
Lemma 7.6, Lemma 7.7 Proposition 7.9	curvature_Ioo_limit, firstCurvature_of_Ioo, firstCurvature_of_polygon_errors (Curvature) sampled_grid_error_le, diameter_le_box, firstCurvature_of_maximal_positive, secondCurvature_of_mirror_first, curvature_of_maximal_positive (Curvature)
Lemma 7.10, Lemma 7.11, Proposition 7.12, Corollary 7.13	injCond1_of_curvature, first_arm_integral_lower, second_arm_integral_lower, lowerSeq_le_of_integral_bounds, arms_strict_of_curvature, injectivity_of_curvature (Curvature); isKi_of_maximal_area (Main)
Lemma 8.1, Lemma 8.2	ki_maximizer_equality_conditions, halfSquareIntegral_combo_gap, displacement_eq0n_of_mamikon_eq, tangentKernel_of_mamikon_eq, middleKernel_of_mamikon_eq (Rigidity)

Result	Lean declarations (continued)
Proposition 8.3, Lemma 8.4, Theorem 8.5	capKernel_of_mamikonS_eq, capKernel_of_triple_midpoint, CapKernel.eq_horizontal_translation, mem_cap_sub_horizontal_iff, sofa_eq_translate_of_upper_support (Rigidity); ki_sofa_eq_gerver_translate, right_angle_monotone_eq_gerver (Main)
Lemma 9.1, Lemma 9.3, Proposition 9.4	eq_of_subset_of_measure_eq, Rigid.recover, Rigid.volume_image, Rigid.isClosed_image (Rigid); regularClosed_cap_sdiff_envelope, isCompact_envUnder, envelope_bounds_of_path_height, envelope_endpoint_order, gerver_regularClosed (RegularClosed)
Fact 2.17	GerverParams.path_snd_lt_one (RegularClosed); gb_x_pos (Gerver/NicheBounds)
Fact 2.1	proposition2_1_2, theorem5_2_2, sigma_Ioc, sigma_periodic (Basic/SurfaceArea)
Fact 2.3, Fact 2.4, Lemma 2.5, Lemma 2.6	theorem1_5_1; proposition2_3_1_exists, proposition2_3_1_subset, theorem2_3_2, lemma2_3_5_supp, theorem2_4_1, theorem2_4_2, theorem2_4_3, theorem2_5_9, theorem2_5_10, proposition2_5_4_isCap, proposition2_5_4_supp, proposition2_5_4_hallway, proposition2_5_4_sets, proposition2_5_4_sigma, sofaArea_mirror (Curvature); IsCap.zero_mem, IsCap.oPt_mem (for Lemma 2.5); IsCap.cPlus_eq_corner, IsCap.aMinus_eq_corner (for Lemma 2.6)
Fact 2.8, Fact 2.9, Fact 2.10, Fact 2.11	proposition3_2_1, proposition3_2_2, theorem3_2_3; lemma3_4_2; lemma3_4_5_one, lemma3_4_5_two, mpc_walk_total, mpc_sum_sigma_sin, mpc_sum_tau_sin; lemma3_4_7, lemma3_4_8, proposition3_3_7, theorem3_3_6, theorem3_1_2
Fact 2.13, Fact 2.14	theorem6_2_3_right, theorem6_2_3_left, theorem6_2_5, proposition6_2_2, proposition6_4_5, proposition6_4_6_deriv, proposition6_4_6_continuous; lemma6_4_2, lemma6_5_5
Fact 2.15	theorem8_1_8, theorem8_2_4, lemma8_3_3, lemma8_3_4, lemma8_3_7, theorem8_3_1, theorem8_3_2, theorem8_3_8, corollary8_5_8, theorem8_4_6, theorem7_1_2_sigma, theorem7_1_2_supp, theorem7_1_2_vertices, theorem7_4_1, theorem7_4_2
Fact 2.16	definition8_1_2_exists, definition8_1_2_unique, Baek.gerver_sofa_area, gerverSofa_area_mem, theorem8_4_1_monotone, theorem8_4_1_walls, theorem8_4_1_tangents, theorem8_4_1_niche (the boundary curves and the area), gn_niche_eq, env_niche_eq (the exact description of the niche), the strict monotonicity lemmas for the abscissa along D , x and B (Gerver/Envelope), theorem6_1_2

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